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MILITARY SPECIFICATION

TRUCKS, FIRE FIGHTING: STRUCTURAL, 1000 GPM

This limited coordinated military specification has beer. prepared by the U.S. Army Belvoir Research and Development Center. based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of MIL-T-10224. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

1. SCOPE

1.1 Scope. This specification covers two types of 1000 gallon per minute (gpm) fire fighting trucks.

1.2 Classification. Fire fighting trucks shall be of the following types, as specified (see 6.2):

Type I fire fighting truck – Without water tower.

Type II fire fighting truck - With hydraulic water tower.

Beneficial comments (recommendations, additions. deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBEDS, Fort Fort Belvoir, 22060-5606 by using the self-addressed Standardization document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4210

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2 .1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifictions and standards (D0DISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-P-35	- Passivation Treatment for Corrosion-Resisting Steel.
RR-C-271	- Chains and Attachments, Welded and Weldless.
VV-F-800	- Fuel Oil, Diesel.

MILITARY

MIL-V-173	- Varnish, Moisture-and-Fungus Resistant (For Treatment of Communications, Electronics and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-1283	- Can, Gasoline, Military, 5–Gallon.
MTL-L-2104	- Lubricatiing Oil, Internal Combustion Engine , Tactical Service.
MIL-G-3859	- Grease Guns, Hand-Operated, Lever, Push and Screw Type.
MIL-E-1OO62	- Engine, Preparation for Shipment and Storage of.
MIL-G-20241	- Casket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-F-24385	- Fire. Extinguishing Agent. Aqueous Film Forming Foam, (AFFF), Liquid Concentrate, for Fresh and Sea Water.
MIL-A-46153	- Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package.
MIL-L-46167	- Lubricating Oil. Internal Combustion Engine, Arctic.
MIL-H-52471	- Hose and Hose Assemblies. Rubber; Hydraulic Pressure Type, General Specification for.
MIL-E-52649	- Engine Cold Starting Aids, Ether Fuel Primers.
MIL-F-52723	- Filter Element, Hydraulic, Disposable,
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

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STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal services.
FED-STD-595	- colors.

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-281	- Automobiles, Trucks, Truck Tractors, Trailers and Trailer Dollies; Preservation and Packaging of.
MIL-STD-154	- Standard General Requirements for Electronic Equipment.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1223	- Nontactical Wheeled Vehicles Treatment, Painting. Rustproofing. Identification Marking and Data Plate Standards.
MIL-STD-1410	Methods for Selection of Individual Engines for End Item Application.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474	- Noise Limits for Army Materiel.
MS 24207	- Coupling, Grease Gun; Hydraulic Type Nozzle.
MS 52129	– Connector, Receptacle, Electrical Inter-vehicle Power Cable.
MS 52131	- Connector. Plug, Electrical Intervehicle Power Cable.
MS 500004	- Pad Eye - Vehicular Towing Lug.
MS 500048	- Towbar, Motor Vehicle: Light, Medium and Heavy Duty (for Vehicles 2-1/2 to 50 Tons and Heavier).

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Other Government publications. The following other Government publications form a part of this specification to the extent specified herein.

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DEPARTMENT OF AGRICULTURE

Forest Service Standard 5100-1 - Spark Arresters.

DEPARTMENT OF TRANSPORTATION

Motor Carrier Safety Regulations

(Application for opies should be addressed to the Superintendent of Documents , Government Printing Office, Washington, D.C. 20402.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the document which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

ANSI/ASME	- Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.
ANSI/AWS D1.1	- Structural Welding Code - Steel .
ANSI B93.28	– Calibration of Liquid Automatic Particle Counters Using “AC” Fine Test Dust.
ANSI B93.30M	- Hydraulic Fluid Power - Contamination Analysis Data - Reporting Method.
ANSI Z26.1	- Safety Glazing Materials for Glazing Motor Vehicles Operating on Land Highways.

(Application for copies should be addressed to the American National Standards Institute, Inc. , 1430 Broadway New York,NT 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 153	- Zinc Coating (Hot-Dip) on Iron and Steel Hardware.
A 240	- Heat Resisting Chromium and Chromium-Nickel Stainless Steel Plate, Sheet and Strip for Pressure Vessels.
A 525	- Steel Sheet, Zinc-Coated (Galvanized by the Hot–Dip Process), General Requirements.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street Philadelphia, PA 19103.)

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lNDITSTRIAL FASTENERS INSTITUTE (IFI)

Std. No. 114 - Standard for Break Manndrel Blind Rivets,

(Application for copies should be addressed to the Industrial Fasteners Institute, 1517 Terminal Tower, Cleveland, OH 44113.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 1901 - Standard for Automotive Fire Apparatus.
NFPA 1931 - Standard for Ladders.

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02110.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT

Rule 442 - Useage of Solvents,

(Application for copies should be addressed to the South Coast Air Quality Management District, 9150 Flair Drive, El Monte, CA 91731.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313,)

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UL Standard 822 - Guide for Certification of Fire Department Pumpers

(Application for copies should be addressed to the Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this Specification and the references tired herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The fire fighting truck, hereinafter referred to as "truck". shall be a diesel-powered unit capable of pumping 1000 gpm of water, and carrying not less than 500 gallons (gal) of water antd 60 gal of foam liquid with necessary equipment and passengers. to the scene of a fire at highway speeds,

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Materialsls. Materials s hall be as specified herein, Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3 .3.1 Material deterioration and control. The truck shall be fabricated from compatible materials. inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage environments and fire fighting conditions to which the truck may be exposed.

3.3.1 Dissimilar metals, Dissimilar metals, as defined in MIL-STD-889. shall be electrically insulated from one another to minimize or prevent glavanic corrosion. Insulation may be provided by an insulating barrier such as a coat of corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape confroming to MIL-G-20241. Protection against any glavanic corrosion could also be obtained by exclusion of the electrolyte, if feasible.

3 .3.1.2 Identification of materials and finishes, The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon re - quest, to the contracting officer or designated representative thereof.

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3.3.2 Recovered materials. Recovered materials are defined as those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the truck may be newly fabricated from recovered materials to the maximum extent practicable, provided the truck produced meets the requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the truck.

3.4 Design. The design of the truck shall be as specified in 3.4.1 through 3.4.2.10.

3.4.1 Angles of approach and departure. The design of the truck shall provide angles of approach and departure which are not less than those specified in the National Fire Protection Association (NFPA) Code Number [No.) 1901.

3.4.2 Major truck units. The truck design shall provide a vehicle having a chassis, cab, water tank, foam system. fire pump system, body with integral stowage units. electrical system, audible warning system. handrails and grabrails, and stowed components.

3.4.2.1 Chassis design. The truck design shall provide a chassis of the cab-ahead-of-engine type. The chassis shall by as specified in 3.11.1 through 3.11 1.10.2.

3.4.2.2 Cab design. The truck design shall provide a cab that is a combination driver's compartment and a jump seat area located as a cab-ahead-of-engine installation. The cab shall be as specified in 3 .11.2 through 3.11.2.3.

3.4.2.3 Water tank design. The truck design shall provide a removable water tank located aft of the fire pump system (see 3.4.2.5). The water tank shall be as specified in 3. 11.3 through 3.11.3.4.

3.4.2.4 Foam system design. The truck design shall provide for the inclusion of a mechanical foam system. The foam system shall be as specified in 3.11.4 through 3.11.4.4.

3.4.2.5 Fire pump system design. The truck design shall provide for the inclusion of a fire pump system. The fire pump system shall be as specified in 3.11.5 through 3.11.5.3.6.

3.4.2.6 Truck body design. The truck design shall provide for a truck body that is a composite formed of integral stowage units, supporting members and necessary panels. The truck body shall be as specified in 3.11.6 through 3.11.6.3.4.

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3.4.2.7 Electrical system design. The truck design shall provide for a 12 volt, direct current (Vdc) electrical system with provisions for a 110 volt, alternating current (Vac) inverter system to accommodate a pair of floodlights. The electrical system shall be as specified in 3.11.7 through 3.11.7.5.3.

3.4.2.8 Audible warning system design. The truck design shall provide for an audible warning system. The audible warning system shall be as specified in 3.11.8 through 3.11.8.4.

3.4.2.9 Handrail and grabrail design. The truck design shall provide for handrails and grabrails to assist personnel boarding and riding the truck. Handrails and grabrails shall be as specified in 3.11.9.

3.4.2.10 Stowed component mounting design. The truck design shall provide for the mounting of holders, brackets or mounting, pads required to stow various components required with the truck. The stowed components shall be as specified in 3.11.10 through 3.11.10.6.

3.5 Construction. The truck shall be constructed so that all mating surfaces are securely fastened , and reinforced where necessary, to resist the twisting and bending forces applicable to emergency fire service.

3.5.1 lnterchanngeability All truck parts having the same part number shall be functionally and dimensionally interchangeable. Interchangeable parts are defined as two or more like parts possessing such functional and physical characteristics as to be equivalent in performance and durability, and capable of being exchanged one for the other without alteration of the parts themselves or of adjoinlng parts, except for adjustment, and without selection for fit or performance.

3.5.2 Piping. All threaded connections of the piping for the water tank. foam system, and fire pump system shall conform to FED-STD-H28.

3.5.3 Wiring. All wiring, except for ignition harness, shall be color coded or numbered for identification. Wiring identification shall be consistent throughout the truck. Wiring shall be neatly run, with all excess wire removed Wiring shall be protected by looms or grommets at all points of potential abrasion. Wiring shall be secured to the truck by corrosion-resistant clamps and fasteners. Wiring shall be secured in compact harnesses and attached to the truck at close intervals. Cables and wires shall not be spliced at any point throughout the length of their runs. Wiring shall not be attached to panels or components that require removal during maintenance, except for actual connection to components. Each 12 Vdc electrical circuit shall be protected with a circuit breakage in accordance with SAE J553).

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3.5.4 Mountings. Mountings for holders, brackets, mounting pads, mounted components (lights. horns, etc.), and for handrails and grabrails shall be constructed by a method that includes the fasteners extending entirely through a backing plate, the mounting surface, and a spacer. The area of the backing plate and the spacer shall be equal to or exceed the area of the mounting surface of the device being mounted. Mountings for attached cabinets shall incorporate r enforcement that transfers load stress to body support members, Mountings, fasteners , backing plates and spacers shall be corrosion resistant metal or hoc-clipped galvanized in accordance with ASTM A 153.

3.5.5 Attaching chains. When components such as caps or plugs for inlets, outlets or connections are specified herein as chain attached to the truck, the attaching chain shall be in accordance with RR-C-271, type II, class 6, either corrosion resistant metal or zinc coated. The chain shall be secured to the truck by clevis attachment. To a mounting pad and by clevis attachment to the cap or plug. The breaking strength of the clevises shall be not less than the breaking strength of the chain. The clevises and mounting pads shall be corrosion resistant metal or hot-dipped galvanized in accordance with ASTM A 153.

3.6 Maintainabillty. Provisions shall be made for adjustment, servicing or replacement of parts and components of the truck. All major components, such as the water and foam liquid tanks, fire pump, transmission. foam proportioner, and booster hose reel when applicable, shall be individually removable and replaceable without disassembly or removal of any other component of the truck. When access openings are used to facilitate adjustment, servicing, or replacement of parts and components, the dimensions of hand or arm access shall be in accordance with SAE J925 for arctic clothing. The edges of access openings shall be rounded and smooth to prevent personal injury. Hinged covers or removable access plates shall be provided for any access opening that will admit water, dust or any other deleterious material during operations. Hinged covers shall be provided with quick disconnect, positive locks and access cover plates shall be provided with quick disconnect fasteners. Scheduled maintenance shall be required not more often than every 50 hours of engine running time, except for before-and-after operation servicing. Each operation in the following maintenance operation list shall be accomplished by one man, in not more than the time specified, using common tools.

a. Remove, replace and adjust all engine-driven belts - 3/4 hour.

b. Remove and replace alternator - 1/2 hour.

c. Remove and replace engine coolant-system hoses - 1 hour.

d. Drain engine lubricating oil. remove and replace oil filter element(s), and refill crankcase - 1 hour.

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e. Remove and replace fuel filter elements - 1/2 hour.

f. Disconnect battery cables, remove and replace batteries, and reconnect battery cables - 1/2 hour.

g. Drain torque converter oil and transmission oil, remove and replace all transmission filter elements and strainers, and refill torque converter and transmission - 1 hour.

h. Remove and replace starter - 1 hour.

i. On the type II truck, remove and replace all hydraulic system filters, screens and strainers - 1 hour,

3.7 Performance. The truck shall perform as specified in 3.7.1 and 3.7.2.

3.7.1 Mobility. The truck shall conform to all performance and mobility requirements of NFPA Code No. 1901. when loaded as follows:

a. Water tank filled with 500 gal of water.

b. Foam liquid tank filled with 60 gal of foam liquid.

c. Fuel tank filled with diesel fuel.

d. All lubricant and hydraulic reservoirs filled to recommended capacity with appropriate fluids.

e. An evenly distributed simulated load of 1500 pounds (lb).

f. A crew of 5 men, each weighting 175 lbs, or an equivalent deadweight substitute for any of them.

3.7.2 Pumping capacity. The truck shall be capable of delivering 1000 gpm of water, in an ambient temperature (see 6.4.1) +125°F, at any altitude from sea level to 2000 feet (ft) above sea level, For the purpose of this requirement, net pump pressure shall be calculated in accordance with NFPA Code No. 1901.

3.8 Human factors. The truck shall conform to the human factors requirements specified in 2.8.1 through 3.8.2,2,

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3.8.1 Operator characteristics. The operator characteristics of the driver's compartment (see 3.11.2.1), the pump operator's panel (see 3,11.6.3), and when specified, the control station for the water tower (see 3.11.12.7), shall provide for operation by personnel ranging from the small person clothed through the large person arctic clothed, in accordance with SAE J833 and SAE J925.

3.8.2 Labeling and marking.

3..8.2.1 Control labels. The control labels for gages, instruments, switches and control handles shall be in accordance with the labeling requirements of MIL-STD-1472. Universal symbology shall be applied to driver's gages, instruments and controls on the dash in accordance with SAE J1500. All valves shall be labeled with a double ended arrow indicating direction of operation for open and close, and their function (i.e. ; foam discharge, water discharge, etc.).

3.8.2.2 Identification and instruction plates. The identification plate (see 3.18) and instruction plates required shall be located on the truck on the flattest, most uncluttered surface available, in such a position that they are not obscured by adjacent items, so as to minimize obscurement by grease, grime, and dirt, and to preclude accidental removal, obstruction or handling damage. Instruction plates shall conform to MIL-P-514, type III, composition C, of type I, grade A, class 1 material, size as applicable. All plates shall be attached by screws, bolts, or rivets,

3.9 Safety. The truck shall conform to the safety requirements specified in 3.9.1 through 3.9.4.

3.9.1 Exterior noise. The exterior sound level of the truck shall not exceed 83 dB(A) when measured in accordance with SAE J366.

3.9.2 Personnel noise. The sound level at the pump operator's position, and any other occasionally occupied position, when the truck is pumping at rated capacity, shall not exceed 85 dB(A) as specified for category D of MIL-STD-1474 when tested as specified in 4.5.2.4.

3.9.3 Personnel protection. All rotating or reciprocating parts and other parts subject to high operational temperatures that are of such nature or are so located as to be a hazard to operating personnel, shall be guarded or insulated to the extent necessary to eliminate the hazard. Protective devices shall not impair the operating efficiency of the truck.

3.9.4 Boarding and riding protection. All walking surfaces and treadplates shall be provided an anti-skid surface. All platforms, steps and handholds shall permit unhampered and nonhazardous boarding, riding and departure from the truck. Hose rollers shall not serve as handrails or grabrails.

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3.10 Environmental requirements.

3..10.1 Altitude pumping, When specified (see 6.2), the truck shall be capable of delivering its rated pumping capacity at the specified service altitude in an ambient temperature of 75° F, +lO°F. Net pump pressure shall be calculated in accordance with NFPA Code. 1901. The contractor may install modification kit(s) for truck service at any altitude above 2000 ft above sea level to conform to this requirement.

3.10.2 Cold starting and operating. The truck shall be capable of starting and operating at -250 F, after the truck has been cold-soaked for not less than 24 hours at an ambient temperature of -25° F. A built-in, ether-priming , cold-starting-aid may be installed to conform to this requirement. The cold starting aid, if installed, shall conform to MIL-E-52649.

3.10.3 Hot starting. The truck shall be capable of starting and operating at +125°F ambient air temperature when the temperature of the engine coolant is at the engine manufacturer 's recommended operating temperature.

3.11 Detailed requirements.

3.11.1 Chassis. The truck shall have a commercial, 4 by 2, front-wheel-steer, rear-wheel-drive type chassis.

3.11.1.1 Frame . The chassis frame shall be a heavy-duty design for fire service use and shall have crossmembers, exlusive of engine supports, to support the gross weight of the cab, body, powerplant, pump, water tank, all other equipment and accessories, and the load specified in 3.7.1, without evidence of distortion that will adversely affect operations, No alteration of the frame shall be made which will reduce its strength.

3.11.12 Engine. For the purpose of selecting the engine, the truck shall be classified as a class II end item in accordance with MTL-STD-141O. The engine, including all systems, components, and accessories, shall conform to MIL-STD-1410, except as otherwise specified herein. All engine accessories shall be furnished or approved by the engine manufacturer for fire service application. The engine shall conform to all operational requirements and conditions specified in table I of MIL-STD-1410, or the engine manufacturer's recommended operating limits , when operating in a +125°F ambient air temperature. The engine shall be liquid-cooled diesel type and shall have not less than 6 cylinders with replaceable liners. The engine shall be furnished with a governor that is set to limit engine speed to the revolutions per minute (rpm) recommended! by the engine manufacturer.

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3.11.1.2.1 Engine horsepower. The engine shall develop not less than 280 brake horsepower (bhp), at an engine speed of not more than 2000 rpm, when operating on diesel fuel conforming to VV-F-800 and lubricating oil conforming to MIL-L-2104 and MIL-L-46167. The power and speed rating of the engine shall be such, that when the truck is operated as specified herein, it shall require not more than) the Environmental Protection Agency (EPA) certified horsepower for that engine.

3.11.1.2.1.1 1 BHP certification. The contractor shall have available a net bhp certification for the engine provided in the truck.

3.11. 1.2.2 Engine lubrication. The engine lubrication system shall be the engine manufacturer 's standard type and shall include an oil filter. Only heavy-duty, full-flow oil filter(s) shall be utilized in the engine lubrication system, and shall be accessible without removing other components. Oil lines, when applicable, to and from the oil filter(s) shall be installed to minimize the effect of vibration and shall be of flexible hose in accordance with the require--ments of SAE J30(d), with reuseable fittings,

3. 11.1.2.3 Engine fuel system. The capacity of the engine fuel tank shall be sufficient to operate the fire pump at rated capacity for not less than 4 hours without refueling, but shall be not less than 50 gal. The fuel tank shall be located in a manner which will not allow spills or overflows to run onto engine, exhaust system, or electrical equipment. The fuel system filler opening shall permit refueling from the spout of a military 5 gal liquid container in accordance with MIL-C-1283. Fuel lines shall be adequately supported and shall be protected against mechanical damage. Construction and mounting shall be such that neither vibration nor tightening of mounting devices will cause leaks to develop. The bottom of the fuel tank and its supporting members shall be not lower than the lowest adjacent part of the chassis or body. The filler cap shall not protrude beyond the body.

3.11.1.2.4 Engine cooling system. The engine cooling system shall be of the capacity recommended by the engine manufacturer. Draincocks shall be provided at the low points in the cooling system and at such other points necessary to drain the cooling system. The draincocks shall be of the hand-operated type that will remain closed under the most severe operating conditions encountered by the truck. The engine cooling system may be equipped with a supplemental heat exchanger . When provided, the heat exchanger shall admit water from the fire pump discharge manifold into the heat-exchanger jacket for cooling, engine coolant without admixture. Engine coolant hoses shall conform to SAE J20. heavy duty type. Engine coolant hose clamps shall be of the double wrap, stainless steel type in accordance with SAE J536, type F, style 1,, 2 or 4. Gaskets, seals, and packing used in the cooling system shall be manufactured of materials capable of resisting deterioration from exposure to mixtures of water and ethylene-glycol based antifreeze conforming to MIL-A-46153.

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3.11.1.2.5 Engine starting system. The engine starting system shall prevent engine start unless the transmission is in a neutral position. The ignition system shall be equipped with a keyless switch and a pilot light.

3.11,1.2.6 Engine exhaust system. The exhaust system sshall include a muffler and tailpipe. The exhaust system be located so that entrance of exhaust gases into the driver's compartment shall be minimized under all conditions of operation. Exhaust and other discharges from the truck shall be directed so that they shall not endanger personnel or obstruct the view of the driver or fire pump operator, Location of the the tailpipe shall prevent heating of tires, disturbance of road dust, and any clogging by mud or water. The muffler shall be located not closer than 2 ft from the engine fuel tank, Where this is not possible, fuel shielding shall be used to protect the engine fuel tank from any part of the exhaust, system, and insulation and shielding elsewhere. The exhaust system shall be provided with a spark arrestor conforming to Forest Service Standard 5100–1 and SAE J350 far not less than 80 percent efficiency. The spark arrestor shall be provided with a means for cleaning out trapped particles. The exhaust system shall not constitute a personnel contact hazard.

3.11.1.3 Drive assembly. The entire drive assembly between the engine and the drive wheels shall have a torque capacity that is not less than the maximum torque developed by the engine through all intervening speed and torque ratios.

3.11.1.3.1 Transmission. The transmission shall be the full automatic type (see 6.4.2), with four forward and one reverse speed. The transmission shall be rated for heavy-duty fire service and designed to match the engine torque to load demand. The transmission shall be protected so that the driver cannot inadvertently shift into reverse, When the transmission is used to drive the fire pump or hydraulic pumps (see 3.11 1,12.1.3), a locking device shall insure that the transmission and its shift lever remains in neutral.

3.11.1.3.2 Drive axle. The drive axle shall be of the full-floating type.

3.11.1.4 Suspension system.

3.11.1.4.1 Springs. Front and rear springs shall sustain the distributed gross vehicle weight (gvw) without evidence of overload or permanent set . The springs shall be designed and suspended for the superimposed load so as to absorb shock and limit sway. Individual wheel loadings, with the truck loaded as specified in 3.7.1, shall be not greater than the rated capacities of the respective springs measured ar the ground.

3.11.1.4.2 Shock absorbers. Shock absorbers shall be the double-acting, hydraulic type, which are matched to the suspension capacity, and are suitable for the gross axle weight rating (gawr).

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3.11.1.5 Wheels and tires. Wheels and tires shall conform to the Tire and Rim Association (TRA) requirements for 50 to 60 mile per hour (mph) speeds for vehicles of the truck's gvw.

3. 11.1.5.1 Front wheels. The front wheels shall be ten hole disk wheels. The front wheels shall be interchangeable with the rear wheels. Wheels shall be inherently in balance.

3.11. 1.5.2 Rear wheels. The rear wheels shall be ten hole disk wheels and shall be interchangeable with the front wheels. Wheels shall be inherently in balance.

3,11 .1.5.3 Tires. Tires shall be not less than 100 level quality. Front tires shall be sized for the load imposed and shall have a highway tread. Rear tires shall be sized for the load imposed and shall have a traction lug tread.

3. 11.1.5.3.1 Tire valves, caps and cores. Tire valves shall clear both wheels and brakedrums. The valves shall be furnished with caps and heat-resistant TRA G-4 cores.

3.11.1.5.4 Spare. A spare wheel and tire shall be furnished and shipped as an overpack, secured in the large–hose bin (see 3.11.6.1.2).

3.11.1.6 Brakes. All brakes shall conform to the latest Department of Transportation (DOT) Motor Carrier Regulation requirements and the provisions of 3. 11.1.6.1 through 3.11.1.6.3.

3. 11.1.6.1 Service brakes, The service brakes shall be straight–air, 4-wheel, self-engergizing type, which shall not require move that 75 lb force for full applications. The service brakes shall control and hold the truck, when loaded as specified in 3.7.1 , on a 30 percent grade. The service brakes shall stop the truck within 30 ft, when loaded as specified in 3.7.1, from a speed of 20 miles per hour (mph) on dry, hard, level surfaces, free of loose material. The service brake mechanisms shall be accessible for exterior adjustment. Brake lines shall be anchored to the chassis and protected from damage. provisions Lo prevent system freezing shall be incorporated in the service brake system. Non-hydroscopic brake fluid shall be used.

3.11.1.6.2 Parking brake. The parking brake shall be all-wheel locking. The parking brake shall be the self-energizing, air-relaese type. The parking brake air system, when charged to not less than 100 lbs per square inch (psi), shall leak not more than 25 percent in 24 hours. The parking brake control shall be of the "push-pull" type, and shall not require more than 50 lb force to fully apply the parking brake. The parking brake, system shall be provided with an emergency brake release mechanism.

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3.11.1.6.3 Engine brake. The engine brake shall be an engine exhaust brake system designed for the engine used in the truck. Braking shall be accomplished by motion transfer through a master-slave piston arrangement which opens the cylinder exhaust valves near the top of the normal compression stroke, releasing t-he compressed cylinder charge to exhaust. The engine brake control shall be an "on-off" switch that permits the engine brake system to be energized by removing pressure from the accelerator pedal when the switch is in the “on” position.

3. 11.1.7 Generating system. The truck shall be equipped with an engine-driven, 12 Vdc alternator/voltage regulator. The alternator/voltage regulator shall have an output rating of not less than 145 amperes (amps). The pulley ratio driving the alternator/voltage regulator shall be sized to provide:

a. Not less than 70 amps at 1000 engine rpm.

b. Not less than 2000 watts (W) of 110 Vac from the inverter (see. 3.11.2.1.7) when the engine is operating at the engine manufacturer’s recommended idle speed.

3. 11.1.8 Steering system. The truck shall be equipped with a power-assisted, hydraulic type. steering mechanism, which shall not require more than 50 lbs turning force to steer the rolling vehicle under normal or power-assist failure operation. The steering mechanism shall preserve , to the maximum extent practicable, the features of directional stability inherent in the manually operated steering mechanism and shall be effective at all vehicle speeds. The mechanism shall permit manual steering in the event of failure of the hydraulic system. The steering mechanism shall include a means for rapid adjustment of lost motion. The mechanism shall provide a cramping angle of the front wheels of not less than 28° right and left at the wheel on the inside of the turning circle. The steering wheel shall be mounted on a rigid and sturdy cloumn. The steering wheel shall be ample size, but shall not interfere with any other projecting equipment in the driver’s compartment or operation of the vehicle.

3.11.1.9 Platform and bumpers.

3. 11.1.9.1 Rear platform. The truck shall be equipped with a rear riding platform. The outer perimeter of the platform shall be of sufficiently heavy construction to serve as a rear bumper, The rear platform height, between the platform riding surface and level ground, when the truck is loaded as specified in 3.7.1. shall be not more than 25.5 inches.

3.11.1,9.2 Front bumper. The front bumper shall be a heavy duty unit attached directly to the frame.

3.11.1.10 Towing hook and towing lugs. A towing hook and towing lugs shall be provided on the truck. The location of attachment shall permit use of a towbar conforming to MS500048, medium duty, but shall not reduce the angles of approach and departure.

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3.11.1.10 Towing hook. A rear mounted towing hook shall be installed on the center line of the truck not less than 6 inches below the bottom of the rear platform. 'The towing hook shall be attached to the frame.

3.11.1.10.2 Towing lugs. Two vehicular towing lugs conforming to MS500004, 10 to 50 ton, shall be installed on the front of the truck. The towing lugs shall be attached directly to the frame below the front bumper.

3.11.2 Cab. The cab S-nail have a closed driver's compartment forward and a jump seat area aft. The jump seat area shall have a roof and sides that extend equally toward the rear of the truck to protect the jump seats, but shall be open to the rear. The partition between the driver's compartment and the jump seat area shall have a sliding glass window not less than 12 inches wide by 6 inches high. Front wheel housings of the cab shall be adequate to permit steering as specified herein. The cab shall conform with the heating ventilating, and air conditioning requirements of SAE J1129.

3.11.2.1 Driver's compartment. The driver's compartment shall be separated from the engine by an insulated fire wall. The interior of the compartment shall have a headliner or false roof, finish trim on exposed vertical panels, and heavy-duty rubber mats on the floor. All wiring and tubing to exterior roof–mounted lights and horns, interior lights, and other accessories shall be run behind the headliner and vertical panels. The compartment interior shall be fitted with insulation behind the headliner, vertical panels, and floor mats to retard heat loss and noise intrusion. All compartment openings shall be sealed with rubber boots, elastomer grommets, elastomer seals, weather stripping, or body compound.

3.11.2.1.1 Driver compartment seats. There shall be two seats provided in the driver compartment. The seats shall have upholstered seats and backs. The cushion material shall be foam rubber or polyurethane, covered with slip-coated vinyl upholstery having a minimum weight of 45 ounces (oz) per linear yard. The driver's seat shall be adjustable fore and aft, and up and down. The passenger seat may be fixed or adjustable. The drivers minimum scat dimensions shall conform to SAE J899.

3.11.2.1.2 Driver's dashboard. The driver's dashboard shall extend across the full width of the driver's compartment. The dashboard shall be installed at such an angle, or provided with an upper shroud, so that any dash lighting shall not reflect in the windshield, or shine directly in the driver's or passenger's eyes. Depth of the dashboard shall be sufficient to mount all of the gages, instruments, switches, monitor lights, controls and compartments specified herein.

3.11.2.1.2.1 Driver's gages and instruments. The driver's gages and instruments required to monitor truck operation and performance shall be in accordance with SAE J209 and shall be installed in the dashboard. The installation of the gages and instruments shall facilitate removal and servicing. The gages and instruments for monitoring truck operation and performance shall include as a minimum, but shall not be limited to, the following:

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a. Speedometer with recording odometer.

b. Tachometer with 0 to 4000 rpm range.

c. Engine oil pressure gage.

d. Brake air tank presssure gage.

e. Engine coolant temperature gage.

f. Transmission oil temperature gage.

g. Ammeter.

h. Fuel gage.

i. Engine hourmeter which indicates actual engine operating hours up to 9999.9 hours, in tenths of an hour.

The gages and instruments shall be individually lighted for night operation. Actuation of night lighting shall be controlled by the headlight switch. The illumination of driver's gauge/instruments shall be rheostat controlled and provide a display luminance of 0.1 to 3.5 cd/m^2.

3.11.2.1.2.2 Driver's operating control s. The driver’s operating controls shall be installed in in the driver's compartment in accordance with the controls requirements of MIL-STD-1472. Controls shall be installed in accordance with SAE J1138 and SAE J898, as follows:

a. Accelerator pedals, transmission shift control pedestal, and service brake pedal on the floor.

b. Ignition switch, parking brake control, engine brake switch and control for parking brake emergency release mechanism on the dashboard.

c. Lighting switches for truck operating lights (see 3.11.7.4.1) and equipment operating lights (see 3.11.7.4.2) requiring driver control on the dashboard.

3.11.2.1.2. 3 Gage and instrument monitors. Remote indicating lights shall be furnished to provide driver warning of the conditions specified in a. through d. below. The indicating lights shall be red and shall light when the condition specified occurs. Means shall be provided to check operation of the lights. In addition, an audible alarm which exceeds operating noise levels by 20 dB(A) shall sound when the condition specified in e. occurs.

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a. When the engine ignition is “on” and the alternator is not charging.

b. When the engine ignition is “on” and the oil pressure is below the engine manufacturer’s recommended minimum.

c. When the engine ignition is “on” and the engine coolant temperature is above the engine manufacturer's recommended maximum.

d. When the engine ignition is “on” and the transmission fluid temperature is above the transmission manufacturer’s recommended maximum.

e. When the air pressure available is less than required for proper brake operation.

3.11.2.1.3 Glass and glazing. Safety glazing of the windshield, door windows, side windows , and rear window shall be provided. Glazing shall conform to ANSI 226.1, Type AS1 for the windshield and Type AS1 or AS2 for other windows. Glass shall be all-tinted. Fixed or hinged glazing shall be sealed in waterproof. stress-absorbent moldings. Moveable windows shall be sealed in extra–heavy-duty, pile-lined channels.

3.11.2.1.4 Windshield wiping and cleaning system. Windshield wipers shall be mounted to facilitate clear vision in accordance with SAE J198. Wiper assemblies shall include means of wiper speed control and return of wiper blades and arms to a retracted position out of the line of driver vision when not in use. The windshield cleaning system shall include a reservoir and a device that will spray the path of the wiper blades with water or cleansing solution. The reservoir shall have a capacity of not less than 2 quarts. The windshield wiping and cleaning system shall conform to SAE J198.

3.11.2.1.5 Heater. A recirculating hot-air heater. which uses engine coolant as a heat source, shall be installed in the driver’s compartment. The heater discharge outlets and ducts shall direct heated air to the compartment floor and to windshield defroster louvers in the top of the dashboard, in accordance with SAE J1129 and SAE J382. The heater shall be equipped with temperature, blower and duct controls that are dashboard mounted.

3.11.2.1.6 Transmission shift control pedestal. The transmission shift control pedestal shall incorporate positive indicators on a quadrant for each gear or range selected. Indicators shall be pointer-to-character, with night lighting of characters on the quadrant. The pedestal shall be located to the right of the driver's seat, in a position that will not interfere with inverter installation between the seats.

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3.11.2.1.7 Inverter. A 6000 W power inverter shall be installed in the driver's compartment. The inverter shall produce 110 Vac at 60 hertz (Hz), ± 1 Hz, and be capable of supplying 6,000 watts into an inductive load such as electric motors. The inverter shall have a voltage control of ± 5 percent, from no-load to full load. The inverter shall be capable of continuous operation at full output. The inverter shall be equipped with an "on-off" switch and a voltmeter which reads output voltage. The inverter shall be located between the seats in the driver's compartment in a position that will provide access to the switch, and visibility of the voltmeter, by both driver and passenger. A remote "on-off" switch and voltmeter for the inverter shall be as specified in 3.11.6.3.4.

3.11.2.1.8 Communications compartment. There shall be a space provided in the driver compartment for installation of communication equipment. The communication compartment space shall be located in a position that will permit either driver or front passenger operation of the communication equipment. The size of the communication compartment shall be 1.5 ft wide by 1.0 ft high by 1.5 ft deep. When specified (see 6.2), a radio communication system shall be provided and the communication compartment may be sized to adapt to the radio communication system equipment provided.

3.11.2.1.9 Driver compartment accessories. The driver compartment shall be equipped with the accessories specified in 3.11.2.1.9.1 through 3.11.2.1.9.4.

3.11.2.1.9.1 Rear-view mirrors. West-coast type outside rear-view mirrors shall be mounted on each side of the cab. The mirrors shall be combination straight and convex type. Stick-on convex mirrors are not acceptable. The mirrors shall be mounted on fold-back support arms and in a position that will allow the driver to survey both sides of the truck. Mirror vision shall not be impaired by truck operation.

3.11.2.1.9.2 Seat belts. Adjustable, weather-resistant, safety seat belts, in accordance with SAE J117, type I, with seat belt anchors in accordance with SAE J787, shall be furnished in the driver compartment for both driver and front passenger. The seat belts shall have non-corrosive, metal-to-metal, quick-release buckles and seat belt retractors.

3.11.2.1.9.3 Interior lights. Interior lighting in the driver compartment shall be provided by a dome light mounted in the headliner or false roof. The dome light shall have an "on-off" control switch. The dome light shall be wired to door switches that will cause the dome light to activate when either door is opened.

3.11.2.1.9.4 Sun visors. Dual sun visors, for the driver and front passenger, shall be provided in the driver compartment.

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3.11.2.2 Jump seat area. The jump seat area of the cab shall be equipped with two jump seats separated by the engine cover.

3.11.2.2.1 Engine cover. The engine cover shall extend from the rear partition of the driver compartment to the front of the body. The engine cover shall be equipped with ventilating louvers. The ventilating louvers shall direct engine compartment air away from the occupants of the jump seats. The engine cover shall be removable for servicing the engine.

3.11.2.2.2 Jump seats. The jump seats shall be located on either side of the engine cover, The space for each jump seat shall be not less than 22 inches wide . A step, or steps, shall be provided for passenger access to the jump seats. The seating portion of the jump seats shall be not less than 22 inches wide. The cushion material shall be foam rubber or polyurethane, covered with slip-coated vinyl upholstery having a minimum weight of 45 oz per linear yard. The back of each seat shall have a padded, upholstered, quick-release type, adjustable bracket capable of holding a complete 30 minute breathing apparatus. With a complete 30 minute breathing apparatus mounted in the bracket, the flat useable seating area of the -jump seats shall be not less than 16 inches from front to rear.

3.11.2.2.2.1 Jump seat safety bars. A moveable safety bar shall be installed across each jump seat. The safety bar shall be 1.25 inch diameter (dia) heavy tubing mounted on a swivel bracket on one side and capable of being positively locked in a quick-release, hand-actuable connector to the right of the seated person.

3.11.2.2.2.2 Dome light. A dome light, with an “on-off” control switch shall be provided in the jump seat area.

3.11.2.3 Soft suction hose compartment. A metal compartment of sufficient size to stow a 12 ft long section of 6 inch dia soft suction hose (see 3.11.10.1) shall be installed at the front of the truck, behind the bumper and in front of the cab. The compartment interior shall be free from projecting nuts or reinforcing brackets. Exposed bolt or rivet heads in the interior of the compartment shall be of the oval type. The compartment shall be equipped with drain holes and shall be finished inside to avoid chafing of the hose.

3.11.3 Water tank. The water tank shall be chassis-mounted and independent of the truck body.

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3.11.3.1 Water tank size. The width of the water tank shall be not more than the width of the large-hose bin of the body (see 3.11.6.1.2). The capacity of the water tank shall be not less than 500 gal. The shape of the water tank and the position of the water tank outlet shall provided that water can be pumped from the tank when the truck is standing in any plane on a 20 percent grade. The interior of the water tank shall be baffled in accordance with NFPA Code No. 1901. The entire water tank top shall be removable. The water tank top shall be gasketed and retained in place, to prevent spillage, by capscrews or similar positive fasteners that do not require special tools to tighten or remove.

3.11.3.2 Water tank structure. The water tank, top and tank mountings shall be manufactured of welded stainless steel conforming to ASTM A 240, Type S30403. After manufacture. and before installation of the water tank on the truck frame, the entire tank shall be sandblasted, or subjected to a pickling treatment, that will remove all mill; and weld scale.

3. 11.3.3 Water tank fill and drain. The water tank shall have a round tank fill opening that has an area of not less than 64 square inches, The tank fill opening shall be equipped with a removeable screen. The screen mesh shall have openings not larger than 0.250 inches or less than 0.2 inches. The closure of the fill opening shall be designed so that water shall not leak from the opening, during acceleration, deceleration or other movement of the truck, from a full or partially filled water tank. The water tank shall be equipped with a nonsplash, combination overflow drain and vent pipe. The overflow drain and vent pipe shall extend downward, toward the rear of the truck, and shall be directed inboard to the ground so that rear wheel traction is not affected by water from the drain.

3.11.3.4 Water tank outlet. The water tank outlet shall be sized to permit the fire pump to draft water from the water tank at a rate of not less than 500 gpm.

3.11.4 Foam system. The truck shall be provided with a mechanical foam system. The foam system shall be an around-the-pump type. The foam system shall consist of a foam liquid tank, a foam liquid metering valve or proportioned, fittings, piping, valves and screens as specified herein. The foam system shall provide for the delivery of foam liquid to the fire pump in a volume that will produce an output of 3 to 8 percent foam solution throughout the operating pressure range of the fire pump at discharge capacities up to 1000 gpm. All parts and components of the foam system that will contact liquid foam shall be of corrosion-resistant material. A complete foam system diagram shall be incorporated in any operation or maintenance manuals furnished with the truck.

3.11.4.1 Foam liquid tank. The truck shall be equipped with a removable foam liquid tank. The foam liquid tank shall be mounted above the fire pump.

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3.11.4.1.1 Foam liquid tank size. The foam liquid tank shall have a capacity of not less than 60 useable gal when filled via the opening described in 3.11.4.1.3. The interior of the tank shall be baffled to prevent excessive surging (see 6.4.3) and foaming of the liquid.

3.11.4.1.2 Foam liquid tank structure. The foam liquid tank shall be manufactured of welded stainless steel conforming to ASTM A 240, Type S30403. After manufacture, and before installation in the truck, the entire tank shall be sand-blasted. or subjected to a pickling treatment, that will remove all mill and weld scale, and then thoroughly rinsed and cleaned to remove all debris .

3.11.4.1.3 Foam liquid tank filler opening. The foam liquid tank shall have a filler opening that will allow filling from a 5 gal container. The filler opening shall have a neck that extends into the tank interior to reduce foaming overflow while filling. The filler opening shall be large enough to permit hand cleaning of the inside of the tank. The filler opening shall be equipped with a removeable screen of a mesh size that will prevent entry of foreign matter into the tank. The filler opening shall be equipped with a gasketed, latching-type cover.

3.11.4.1.4 Foam liquid rank vent. The foam liquid tank shall have a splash-proof vent that is piped to the outside of the truck.

3.11.4.1.5 Foam liquid tank outlet. The outlet of the foam liquid tank shall be equipped with a removeable screen to prevent foreign matter or jelled foam liquid from entering the piping and valving of the foam system.

3.11.4.1.6 Foam liquid tank drain. The foam liquid tank shall be equipped with a drain outlet. A drain valve shall be installed at the outlet. A drain pip? shall be attached to the drain valve and shall be located so that the tank is drained completely and is directed down so that the discharge point is below the bottom of the truck frame.

3.11.4.2 Foam liquid metering valve or proportioned. A calibrated foam liquid metering valve or proportioner shall be provided in the foam system to regulate the discharge or proportioning of foam liquid from the foam liquid tank to the fire pump. The foam liquid metering valve or proportioner shall be designed for remote regulation of the foam liquid flow as a percentage of the total solution flow from the fire pump.

3.11.4.3 Foam system piping and valves. A quick-acting shut-off valve shall be provided in the foam system piping between the foam liquid tank and the foam liquid metering valve or proportioner. The shut-off valve shall be as close to the foam liquid tank as possible. A check valve shall be provided between the foam liquid metering valve or proportioned and the eductor at the fire pump suction manifold to prevent water or solution from flowing back from the mani-fold to the liquid foam delivery system.

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3.11.4.3.1 Foam system flushing provisions. A means for flushing the foam piping and valving shall be installed in the foam system. The flushing provisions shall be designed to permit a complete flushing of the system and shall assure that no foam liquid residue will accumulate to clog the system when proper flushing has been performed. Complete flushing and cleaning instructions shall be incorporated in any operation or maintenance manual furnished with the truck.

3.11.4.4 Foam system controls, Controls for operation of the foam system shall be located as specified in 3,11.6.3.1.3, An instruction plate showing the proper settings of the metering valve or proportioned shall be located adjacent to the foam system controls.

3.l1.5 Fire pump system. The fire pump system shall incorporate a fire pump . suction section. discharge section, piping and drains. The fire pump system shall exhibit no evidence of leaking when subjected to the hydrostatic test specified in 4.5.2.11. A complete diagram of the pumping system shall be incorporated in any operating and maintenance manuals furnished with the truck.

3.11.5.1 Fire pump. The fire pump shall be the centrifugal type. When drafting through a 6 inch dia suction hose, the fire pump shall attain a minimum rating of 1000 gpm in accordance with Underwriter's Laboratories (UL) Standard 822. The fire pump shall be equipped with a priming device that conforms to UL Standard 822.

3. 11.5.1.1 Fire pump drive. The fire pump shall be driven by the truck engine, through the transmission to the pump drive. The temperature rise in the gear oil of the pump drive shall not exceed the maximum gear oil temperature recommended by the fire pump manufacturer when pumping in continuous operation. The Pump drive may be equipped with supplemental oil-cooling to assure that fire pump gear oil temperature rise will not exceed the fire pump manufacturer's recommended maximum gear oil temperature for continuous operation. The control of the fire pump drive shall be as specified in 3.11.6.3.1.7.

3.11.5.2 Suction section. The fire pump system suction section shall include a fire pump suction manifold, large suction inlets, small suction inlets, and the necessary piping, valves and connections required to assure the fire pump will draft as specified herein. The suction manifold and piping for the suction section shall be corrosion-resistant metal, passivated in accordance with QQ-P-35, or galvanized inside and out, in accordance with ASTM A 525, G 90 coating.

3.11.5.2.1 Fire pump suction manifold. The fire pump suction manifold shall be providede with the necessary ports, sized to accomodate the flow volumes indicated, for piping attachments as specified herein. The fire pump suction manifold shall direct suction flows to the fire pump inlet,

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3.11.5.2.2 Large suction inlets. Large suction inlets shall be 6 inches dia. There shall be three large suction inlets mounted on each truck. One inlet shall be located at the front of the truck, offset to the curbside and one inlet on each side of the truck. The inlets shall be not more than 4 feet above ground level. Each inlet shall have 6 inch National Hose (NH) male threads (thds) and machined ends for suction hose coupling or valve gasket seating. Each inlet shall be furnished with a readily accessible, removable brass strainer. The inlet at the front of the truck shall be equipped with a valve. The valve shall have a positive, leakproof seal and a hand-adjustable control handle. The front suction inlet and the hose coupling shall not extend past the bumper. The two side inlets shall be furnished with long-handled, gasketed caps. The suction inlets shall be piped to appropriate sized ports in the suction manifold.

3. 11.5.2.3 Small suction intels. Small suction inlets shall be 2.5 inches dia. There shall be two small suction inlets located on the left, or streetside of the truck, at the bottom of the pump operator's panel (see 3.11.6.3). The inlets shall be gated and gasketed female, swivel suction inlets, with internal NH thds. Each inlet shall be furnished with a rocker-lug, male plug. The plug shall be attached to the truck with a chain. The inlets shall be piped to appropriate sized ports in the fire pump suction manifold through 2.5 inch dia, balanced, corrosion-resistant, full flow valves. The valves shall be operated by individual control handles by single-handed operation. Controls for the valves shall be as specified in 3.11.6.3.1,5.

3.11.5.2.4 Water tank suction piping. A quick-opening shut-off valve shall be installed in the piping from the water tank to the fire pump suction manifold. The valve shall be designed for remote opening and closing. A check valve shall also be installed in the piping to prevent any back-flow of water or foam liquid solution from the fire pump from reaching the shut-off valve or the water tank. The piping between the shut-off valve and the fire pump suction manifold may incorporate flexible, noncollapsible connections to absorb vibration. The piping shall have 2 drain valve of not less than 0.75 inch dia installed at its lowest point to permit draining of the piping and check valve while the shut-off valve is closed. The drain valve shall be designed for remote control. A drain pipe from the drain valve shall discharge to the ground at a point not above the bottom of the frame. The piping, valves and connections shall be sized to flow not less than 500 gpm and shall withstand a working pressure of not less than 120 psi. The controls for the shut-off valve and the drain valve shall be as specified in 3.11.6.3.1.4.

3.11.5.2.5 Suction manifold fittings. The fittings that connect piping to the ports of the suction manifold and connect the manifold to the fire pump inlet shall be corrosion-resistant .

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3.11.5.3 Discharge section. The fire pump discharge section shall include a fire pump discharge manifold, a fire pump discharge relief valve, discharge control valves, discharge outlets. piping, connections and discharge outlet drains as specified herein. The discharge manifold and piping in the discharge section shall be corrosion-resistant metal passivated in accordance with QQ-P-35. or galvanized inside and out in accordance with ASTM A 525, G 90 coating.

3.11.5.3.1 Fire pump discharge manifold. The fire pump discharge manifold shall be provided with the necessary discharge ports for attachment of valves and piping as specified herein. The ports shall be sized to accommodate the flow volumes required. The fire pump suction manifold shall distribute discharge flows from the fire pump to the valves and piping.

3.11.5.3.2 Discharge relief valve. The discharge relief valve shall automatically control the discharge pressure according to the setting established by the pump operator. The design of the discharge relief valve shall provide that an increase of not more than 30 psi pump pressure shall occur when: a) the pressure relief setting of the valve is 10 psi higher than the pump working pressure. and b) all discharge valves are slowly closed. The relief valve shall be designed to operate at any fire pump working pressure above 75 psi when the fire pump is pumping at rated capacity. The design of the relief valve shall incorporate provisions for the attachment of an indicator light that will indicate discharge relief valve actuation. The location of the relief valve shall be such that remote adjustment of the discharge relief valve setting can be effected. Relief valve shall discharge away from operator and wheels. Control of the relief valve and location of the indicator light shall be as specified in 3.11.6.3.1.8.

3.11.5.3.3 Discharge control valves. The discharge control valve shall be properly-sized, balanced, corrosion-resistant, full-flow-through type valves designed for remote control. Controls for each discharge control valve shall be as specified 3.11.6.3.1.6. Each control valve shall by labled “discharge” and systematically numbered according to the corresponding discharge outlet and valve control. Numbers shall be affixed by corrosion-resistant, permanently attached tags.

3.11.5.3.4 Discharge outlets. Discharge outlets that have NH thds, or adapters that connect to hoses, shall be of the rocker lug type and shall have machined ends to provide a seat for hose coupling gaskets or cap gaskets. Discharge outlets shall be designed into the truck, as follows:

a. Four 2.5 inch NH discharge outlets; two on the streetside and two on the curbside of the truck. Each outlet shall be equipped with a 2.5 inch 30- or 45-degree elbow. Each elbow shall be equipped with a 2.5 inch to 1.5 inch NH adapter. The adapter shall be capped with a 1.5 inch cap which has an attaching chain.

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b. One discharge outlet, piped into the water tank, sized to flow not less than 50 gpm.

c. Two discharge outlets, sized to flow not less than 500 gpm with an internal pressure drop of not more than 35 psi, at the small-hose bin (see 3.11.6.1.1) located above the fire pump. Each outlet shall be furnished with a swivel and a 1.5 inch NH fitting. The outlet shall be located to allow playout of the hose to either side of the truck.

d. Two National Pipe Thread (NPT) discharge outlets, sized to flow not less than 500 gpm with an internal pressure drop of not more than 35 psi to the rear of the truck at the leader-hose bins (see 3.11.6.1.3). Each outlet shall be furnished with a 2.5 inch NT fitting and a 2.5 inch to 1.5 inch NH adapter.

The discharge outlets shall be systematically numbered according to the corresponding discharge valve and control. Numbers shall be affixed by corrosion-resist ant, permanently attached labels adjacent to the outlet.

3.11.5.3.5 Discharge outlet piping. Each discharge outlet shall be piped to its applicable discharge control valve, using corrosion-resistant connections where necessary. The method of piping shall accommodate the discharge outlet drains specified in 3.11.5.3.6, as applicable.

3.11.5.3.6 Discharge outlet drains. The discharge outlets specified in 3.11.5.3.4a shall have a drain valve of not less than 0.75 inch dia installed at the lowest point in its piping to permit draining while the discharge contrl valve is closed. Drain valves shall be designed for remote control. Drain pipes from the drain valves shall be concealed behind truck body panels and shall discharge to the ground at a point net above the bottom of the frame. Controls for the drain valves shall be as specified in 3.11.6.3.1.9.

3.11.6 Truck body. The truck body shall be chassis-mounted. The body shall provide for wheel housings and running boards. The body shall also provide for the attachment of stowage units (hose bins, cabinets, etc.). The truck body shall be reinforced at all points of attachment.

3.11.6.1 Integral stowage facilities. Integral storage facilities shall be designed into the truck body. The integral stowage facilities shall consist of a small-hose bin, a large-hose bin, two leader hose bins and a through cabinet. Design of the integral stowage facilities shall be as specified in 3.11.6.1.1 through 3.11.6.1.4.3. The intergal stowage facilities shall be labeled to indicate stowed contents in accordance with the labeling require-ments of MIL-STD-1472.

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3.11.6.1.1 Small-hose bin. A small hose stowage bin shall be designed to be located across the top of the truck body, above the fire pump, the suction and discharge manifolds, and the control valving. The small hose bin shall be as long as the full width of the truck body. The small-hose bin shall have a removable metal divider that provides two equal small-hose stowage compartments. The size of the bin shall be large enough so that each compartment shall be capable of holding nor less than 200 ft. of 1.75 inch dia preconnected hose, when the divider is in place. The bottom of the entire bin shall be removable when the divider is removed. The opening created by removal of the bin bottom shall be sufficient to accommodate fire pump removal from the truck when the fire pump is disconnected from the suction and discharge sections. The removable hose bin bottom shall be designed as specified in 3.11.6.1.1.1. The metal bin bottom shall be protected by a hose bin floor as specified in 3.11.6.1.1.2. The interior of the bin shall be free of projecting nuts, sharp angles or reinforcing brackets. Exposed bolt or rivet heads in the bin interior shall be of the oval type. Hose rollers, to serve as guides when playing out hose and to prevent hose chafe, shall be installed on the side and bottom openings of the bin. Hose rollers shall be corrosion-resistant metal that has been passivated in accordance with QQ-P-35. Hose shall not be furnished.

3.11 1.6.1.1.1 Hose bin bottoms. Hose bin bottoms shall be full metal panels with raised crossmembers for support of a hose bin floor. The crossmembers shall have provisions for the attachment of hose bin floors. Bin bottoms shall have drain holes between crossmembers to prevent water retention in the bin. Bin bottoms shall be capable of supporting 200 lbs of dead weight on any 1 square ft of floor area without permanent defoliation or fracture.

3.11.6.11.2 Hose bin floors. Hose bin floors shall be slatted units. Slats shall be seasoned, select-grade hardwood or aluminum. Slats shall be not less than 3.5 inches wide and have a finished thickness of 0.75 inch. The spacing between slats shall be not less than 0.625 inch nor more than 0.75 inch. The slats shall be fastened to hose bin bottom crossmembers by countersunk screws or bolts. The slats of the floor shall be designed for easy removal to permit bin bottom removal.

3.11.6.1.2 Large-hose bin. A large-hose bin shall be the inner top of the truck body, aft of the small-hose bin specified in 3.11.6.1.1. The large-hose bin shall be not wider than the distance between the rear wheel housings and shall extend lengthwise to the end of the truck body. The large-hose bin shall have a removeable metal divider that separates the bin into two equal, lengthwise compartments. The size of the bin shall be large enough so that each compartment shall be capable of holding not less than 1200 ft of 3 inch. double-jacketed. rubber-lined fire hose. The bin bottom shall be as specified in 3. 11.6.1.1.1 and the bin floor shall be as specified in 3.11.6.1.1.2. The water tank fill opening or cover, and the water tank overflow drain and vent shall not extend into the large-hose bin. The interior of the bin shall be free from projecting nuts, sharp angles, or reinforcing brackets. Exposed bolt or rivet heads in the bin interior shall be of the oval type. Hose shall not be furnished, but a fitted cover as specified in 3.11.6.1.2.1 shall be furnished.

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3. 11.6.1.2.1 Fitted cover. A removeable, fitted cover shall be provided for the large-hose bin. The front of the cover (end nearest the cab) shall be provided with matching fasteners for attachment to the truck. The sides shall be fitted with stretch cords and hooks, or an equally effective, flexible attaching method. The design of the cover shall allow playout of hose from either compartment. individually or collectively without damage to the cover or detachment of any side or end fastening. Cover material shall be Herculite or equal.

3.11.6.1.3 Leader-hose bins. Two leader-hose bins shall be designed into the truck body, outboard on either side of the large-hose bin, The size of bins shall be large enough so that each bin is capable of holding not less than 300 ft of 2.5 inch, preconnected fire hose. The bin bottoms shall be as specified in 3.11.6.1.1.1 and the bin floors shall be as specified in 3.11.6.1.1.2. The interior of the bins shall be free from projecting nuts, sharp angles, or reinforcing brackets. Exposed bolt or rivet heads in the interior of the bins shall be of the oval type. Hose shall not be furnished.

3.11.6.1.4 Through cabinet. The through cabinet shall extend completely through the truck body, from curbside to streetside, forward of the fire pump and below the small-hose bin. The through cabinet shall be not less than 8 inches wide and 36 inches high. The through cabinet shall be common to both sides of the truck and shall have doors at either end. The through cabinet shall be equipped with a removable shelf. The shelf shall be located midway between the top and bottom of the cabinet. The removable shelf shall be less than the interior cabinet length to permit mounting of components on at least one door (see 3.11.10.5). The bottom of the cabinet shall be as specified in 3.11.6.1.1.1 for bin bottoms. The cabinet shall be floored as specified in 3.11.6.1.1.2,

3.11.6.1.4.1 Cabinet doors. Cabinet doors shall be of the flush, box-type design and fully gasketed, equipped with spring type, hold-open, door devices. Door hinges shall be corrosion-resistant . full length piano-type. Door handles shall be recessed "D" ring-type, conforming to MIL-STD-1472 requirements for handles and grasp areas, with two-point latching. Door locks shall be flush, bolt-type.

3.11.6.1.4.2 Cabinet illumination. The cabinet shall be illuminated in the interior, Lighting shall be automatically controlled through door switches that activate the lights when the door is opened.

3.11.6.1.4.3 Cabinet ventilation and drain. The cabinet shall be ventilated to prevent interior condensation and have drain holes.

3.11.6.2 Attached equipment cabinets. Equipment cabinets shall be designed to attach to the truck body. Five cabinets shall be located on the streetside of the truck and two cabinets on the curbside of the truck. The streetside cabinets shall be arranged as two lower cabinets and three upper cabinets. The curbside cabinets shall be two lower cabinets only.

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3.11.6.2.1 Streetside cabinets. One lower streetside cabinet shall be provided to the rear of the whneel housing and shall extend as far as practicable to the rear of the truck body. The other lower streetside cabinet shall be provided in front of the wheel housing and shall extend as far forward as practicable, but shall not interfere with the pump operator's functions, when the pump operator is working at the pump operator's panel (see 3.11.6.3). The two lowe streetside cabinets shall be not less than 25 inches high, and shall have door–to-back depth equal to the wheel housing width, but not less than 20 inches. The cabinets shall be recessed not less than 0.75 inch from the outer edge of the running board, Panel fillers shall be provided as necessary, on the sides or tops of the cabinets to dress the wheel housing and cabinet tops flush, The three upper cabinets shall extend the full length of the two lower cabinets and shall be of equal width. The upper cabinet height shall be such that the cabinet tops shall be flush with the (side) tops of the leader-hose bins. The door-to-back interior depth of the upper cabinets shall be sufficient to stow complete 30 minute breathing apparatus in vertical brackets, but not less than 12 inches. The upper cabinets shall be vertically flush with the lower cabinets and any wheel-housing skirts. The upper cabinet tops shall run continuously, without any gaps, and shall conform to 3.9.4. All streetside cabinet bottoms and floors shall be as specified in 3.11.6.1.1.1 and 3.11.6.1.1.2. Cabinet doors, illumination and ventilation shall be as specified in 3.11.6.1.4.1 through 3.11.6.1.4.3.

3.11.6.2.2 Curbside Cahinets. The curbside cabinets shall be identical to. but observe copies of the two lower streetside cabinets. The curbside cabinet tops shall be connected with a filler panel to provide a continuous top, wit-bout any gaps. which shall conform to 3.9.4.

3.11.6.3 Pump operator’s panel. A pump operator‘s panel shall be provided on the streetside of the truck body. The panel shall be located forward of the streetside cabinets and aft of the through cabinet. The pump operator’s panel shall accommodate the streetside inlets and outlets specified herein, and the controls, gages and instruments specified in 3.11.6.3.1 through 3.11.6.3.3.3, The inlets, outlets, controls, gages and instruments on the control operator’s panel shall be labelled in accordance with the Labeling requirements of MIL-STD-1472. The controls shall be in accordance with the Controls requirements of MIL-STD-1472. Controls shall be located no higher than 73 inches and no lower than 36 inches above the ground.

3.11.6.3.1 Pump operator’s controls. The controls mounted on the pump operator‘s panel shall have plates or collars installed at the point where they emerge through the panel. The controls shall be as specified in 3.11.6.3.1.1 through 3.11.6.3.1.9.

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3.11.6.3.1.1 Engine throttle control. The engine throttle control on the pump operator's panel shall be of the screw-locking, center-pushbutton-release type, The throttle control body shall hold the control lever in any desired position and shall not be affected by vibration or the pull of the accelerator spring. The engine throttle shall control engine rpm from manufacturer's recommended idle to maximum.

3.11.6.3.1.2 Heat exchanger control. When a heat exchanger has been provided (see 3.11.1.2.4),-a rotary or linear "on-off" control for the heat exchanger shall be mounted on the pump operator's panel.

3.11.6.3.1.3 Foam liquid control. The foam liquid control on the pump operator's panel shall be a rotary control. A dial and pointer arrangement capable of calibration shall indicate regulation of discharge percentage or proportioning of foam liquid to total solution flow, An instruction plate, with chart, indicating the proper setting shall be located on the pump operator's panel adjacent to the control,

3.11.6.3.1.4 Water tank suction controls. The water tank shut-off valve control on the pump operator's panel shall be a linear control. The drain valve control for draining the water tank suction piping shall be located on the pump operator's panel adjacent to the water tank shut-off valve control.

3.11.6.3.1.5 5 Small suction inlet controls. The controls for the small sucticn inlets shall be mounted on the pump operator's panel adjacent to the inlets. The controls shall be linear.

3.11.6.3.1.6 Discharge valve controls. The controls for the discharge valves shall be linear (push-pull). All of the discharge valve controls shall symetrically grouped. The controls shall be numbered to correspond to the applicable discharge outlet and control valve.

3.11.6.3.1.7 Fire pump drive control. A control for the fire pump arive may be mounted on the pump operator's panel. If a fire pump drive control is not mounted on the pump operator's panel. it shall be mounted on or under the dashboard in the driver's compartment of the cab. The pump drive control shall be linear (push-pull) with positive locking in either position.

3.11.6.3.1.8 Discharge relief valve control. The control for the discharge relief valve on the pump operator's panel shall be a rotary control. A dial and pointer arrangement capable of calibration shall indicate the extent of valve opening. An indicator light shall be connected to the discharge relief valve and mounted on the pump operator's panel adjacent to the discharge refief valve control.

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3.11.6.3.1.9 Discharge outlet drain controls. The controls for the discharge outlet drains shall be mounted on the pump operator's panel beneath the associated discharge valve control.

3.11.6.3.2 Pump operator's gages. The gages mounted on the pump operator's panel shall have plates or collars installed where they emerge through the panel. The gages shall be as specified in 3.11.6.3.2.1 through 3.11.6.3.2.4.

3.11.6.3.2 Master pressure gage. A compound master pressure gage on the pump operator's panel shall have a dial pressure range of 30 inches of mercury (Hg) to a pressure of not less than 600 psi. The dial face of the gage shall be not less than 3.5 inches dia.

3.11.6.3.2.2 Master suction gage. A compound master suction gage on the pump operator's panel shall have a dial pressure range of 30 inches of Hig to a pressure of not less than 600 psi. The dial face of the gage shall be not less than 3.5 inches dia.

3.11.6.3.2.3 Discharge outlet gages. A separate pressure gage shall be provided for each discharge outlet, Each gage shall be provided with *a* device to eliminaate indications of pressure fluctuatuin (glycerin-filled gages are acceptable). The gages shall be not less than 2.5 inches dia. The gages shall be grouped in accordance with MIL-STD-1472 control requirements dirctely above the controls associated with the discharge outlets.

3.11.6.3.2.4 Engine and transmission operating gages. Engine and transmission operating gages shall be provided on the pump operator 's panel, located to the left of the pump control panel and separated from it by not less than 2 inches, as follows:

a. Enginp oil pressure

b. Engine coolant temperature

c. Transmission oil temperature

d. Fuel.

The gages shall be not less than 2.5 inches dia. The gages shall be symmetrically grouped in the proximity of the engine throttle control and pump drive control. Each gage shall be provided with a remote indicating light to monitor conditions as specified in 3.11.2.1.2.3b, c, and d.

3.11.6.3.2.5 Water tank gage. A water tank gage that indicates the water level in the water tank shall be mounted on the pump operator's panel. The gage shall be located 62 inches, +2, -6 inches above ground level.

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3. 11.6.3.3 Pump operator' s instruments. The instruments mounted on the pump operator's panel shall have plates or collars installed where they emerge through the panel. The instruments shall include an engine tachometer. an anmeter and two test connections.

3.11.6.3.3.1 Engine tachometer. An engine tachometer shall be provides on the pump operator's panel. The tachometer shall be identical to and interchangeable with the tachometer specified in 3.11.2.l.2.lb. The tachometer shall be located directly above the engine throttle control. The tachometer shall be equipped with the means for checking the tachometer accuracy with a hand-held revolution counter without disconnecting the tachometer.

3.11.6.3.3.2 Ammeter. An ammeter shall be provided on the pump operator's panel . The ammeter shall be not less than 2.5 inches dia. The ammeter shall be located in proximity of the engine operating gages specified in 3.11.6.3.2.4.

3.11.6.3.3.3 Test connections. Two O .25 inch NPT female test connections far test gage attachment shall be provided on the pump operator's panel. The connections shall be located in close proximity to the master suction and master pressure gages. The connections shall be piped to the respective suction and discharge manifolds and labelled accordingly. The connections shall be plugged with removable NPT male plugs of corrosion-resistant material .

3.11.6.3.4 Remote inverter controls. A duplicate switch which turns the inverter on and off and voltmeter as specified in 3.11.2.1.7, shall be mounted on the forward wall of the through cabinet. The location of these controls shall be such that the pump operator can observe the voltmeter from his normal operating position when the through cabinet door is open.

3.11.7 Electrical system. The electrical system for the truck shall be wired as specified in 3.5.3. Wiring for the 110 Vac inverter system shall be run separate from the 12 Vac wiring. A complete wiring diagram of the entire electrical system shall be incorporated in any operating or maintenance manual furnished with the truck.

3. 11.7.1 Electrical equipment. All electrical equipment of the truck shall be 12 Vdc, except for the inverter, inverter receptacles, the wiring between them, and the 300 W floodlights specified in 3.11.7.4.4. All electrical equipment shall meet the criteria set forth in MIL-STD-454, Requirement 1. All exposed parts which are electrically energized shall be insulated. fully enclosed or guarded. Components shall be located so as snot to be adversalyaffectedbyengine heat. All electrical components shall be water-resistant and designed so that their normal function will nor be impaired by entrance of water during fire fighting operations, heavy rains, from road splash, from condensation. or from water spray resulting from other causes. All noncurrent carrying metal parts shall be grounded. Any 6 volt (V) equipment, paired in series with resistors or tapped to the midpoint of one of the batteries, shall not be used in the truck.

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3.11.7.2 Storage batteries. Two 12 V storage batteries shall be furnished with each truck. The storage batteries shall conform toc SAE Nos. 20T6A or 20T8A. The batteries shall be stowed in a battery tray, or trays, designed to be of sufficient size to hold and store the batteries specified. Batteries shall be shipped with the truck in a dry and charged condition.

3. 11.7.2.1 Battery tray(s). The battery tray(s) may be either draw-out or fixed type. If a draw-out type is provided on the truck, there shall be a means of positive locking of the tray(s) in the retracted position and preventing battery cable chafing provided. If a fixed type is provided on the truck mounting on the streetside running board shall be permitted, provided no part of the tray(s) extend beyond the attached equipment cabinets. Batteries shall be securely held in place by means of a removable locking device and shall be protected against engine and exhaust heat. The battery tray(s) shall be ventilated and treated against acid fumes and spillage. The battery terminals shall be provided with guards to prevent accidental shorting.

3.11.7.3 Battery circuits. Battery circuits shall conform to 3.11.7.3.1. 3.11.7.3.2 and the wiring diagram titled "Suggested Application of Dual Batteries to Single Ignition Engine" in the appendix to NFPA Code No. 1901.

3.11.7.3.1 Battery charging circuits, station. Each battery shall be connected to an electrical receptacle, that conforms to MS32129, for station charging. The receptacle. shall be mounted on the truck in such a location that forward movement of the truck will result in an automatic disconnect without damage to the receptacles aboard the truck or the mating plug and its attached cables from the station charging unit. The mating plug, in accordance with MS52131, shall be furnished with the truck.

3.11.7.3.2 Battery charging circuits, truck. The battery circuits between the generating system see 3.11.1.7) and the batteries shall have reverse polarity devices that protect the alternator/voltage regulator from damage when the batteries are connected in reverse polarity. The battery corciots shall have a control system installed whereby:

a. Either battery may be used for:

   (1) Ignition and lights alone
   (2) Starting alone
   (3) Starting, ignition and lights

b. Both batteries may be used simultaneous]y in engine starting.

c. Neither lighting wiring nor ignition wiring will be connected to both batteries except when both starter switches are operated.

d. The alternator/voltage regulator shall be connected to supply the battery being used for ignition and lights.

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3.11.7.4 Lighting system. The truck lighting system shall be designed to include truck operating lights , equipment operating lights, warning lights and the 110 Vac floodlights.

3.11.7.4.1 Truck operating lights. All truck operating lights. reflectors and wiring shall conform to the DOT Motor Carrier Safety Regulatios, except as otherwise specified herein:

a. DOT identification lamps are required.

b. Clearance and side marker lamps shall be combined and mounted to indicate width of the truck.

c. Rear–mounted lamps shall be located so that they shall not be obscured by equipment or personnel normally carried on the platform.

d. Front turn-signal units its shall be single-faced. flush–mounted. The turn-signal controller shall be the self-cancelling type and shall be mounted on the steering column; shall operate visual indicators on the dash: and shall be wired through the ignition system.

e. Stoplights, tail lights, rear turn-signal and back-up lights shall be flush-mounted on the rear of the body. Back-up lights shall be not less than 5 inches dia and shall automatically be activated when the Transmission is shifted into the reverse mode.

3.11.7.4.2 Equipment operation lights. The truck equipment operation lights shall be as follows:

a. Two automotive-type sealed beam spotlights shall be located on each side of the windshield, with controls extended inside the cab for “on-off” control and movement of the lights through horizontal and vertical angles.

b. Pump operator’s panel lights. enclosed within hoods or covers to cpmpletely illuminate the panel, but prevent glare.

c. Two sealed-beam hose pickup lights. mounted outside the hose bins. Hose pickup lights shall not be mounted on or attached to, any handrail or grabrail.

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d. Step lights, for the cab and jump seat steps controlled by the headlight switch. Step lights shall be hooded or shielded for downward illumination only.

e. Two engine-compartment lights, to illuminate both sides of the engine. Both lights shall be shielded and controlled by one switch on the clash, convenient to the driver.

3.11.7.4.3 Warning lights. Warning lights shall conform to NFFA Code No, 1901. Two motorized, oscillating sine-wave-pattern warning lights shall be mounted beneath the windshield in equally spaced positions on the front of the cab. Two strobe-type warning lights shall be mounted on the rear of the truck, outside the hose pickup lights. All warning lights shall have red lenses.

3.11.7.4.4 Floodlights, The portable, 300 W, 110 Vac floodlights, with 3-wire grounded connectors (Circle D Portable Light No. 150 and model P-23 Holddown Clamps or equal), shall be mounted on either side of the truck above the through cabinet.

3.11.7.4.5 Floodlight receptacles. There shall be four 110 Vac floodlight receptacles mounted on truck body panels. There shall be one receptacle mounted on each side of the truck, adjacent to the fire pump, and two receptacles mounted near the top of the truck body, above the fire pump. The receptacles shall be dual, grounded, and weatherproof, with spring-loaded covers. Wiring between the inverter and the receptacles shall be of sufficient gage, fused, properly supported and protected from injury. Each receptacle shall be labeled 115 Vat, 60 hz, and shall conform to the labeling requirements of MIL-STD-1472.

3.11.7.5 Lighting system controls. Lighting system controls, for light controls not otherwise specified herein, shall be as specified in 3.11.7.5.1 through 3.11.7.5.3. Labels and marking for lighting system controls shall conform to the labeling requirements of MIL-STD-1472.

3. 11.7.5.1 Headlight control. The headlight control shall be a push-pull type switch that shall activate headlights, tail lights, and step lights. Rotation of the headlight switch s'nail dim or brighten the gage and instrument lights specified in 3.11.2. 1.2.1, and shall activate the dome light in the driver's compartment at the end of the rotary dimming revolution. The headlight switch shall be mounted on the dash in the driver's compartment.

3.11.7.5.2 Warning light controls. Warning light switches, that will activate the warning lights specified in 3.117.4.3. and the warning lights specified for each type of truck, (see 3.11.11.5 and 3.11.12.10) shall be single-pole, single-throw, rocker, circuit-breaker type switches, with indicating light for the "on" position. The switches shall be mounted on the dash in the driver's compartment in a group of three adjacent, identical switches, independent of other electrical equipment controls, Shielding of the indicating lights shall be provided if necessary to prevent glare on the windshield.

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3.11.7.5.3 Floodlight receptacle controls. The two upper floodlight receptacles shall be controlled by two "on-off" switches. The switches shall be mounted on the dash in the driver's compartment.

3.11.8 Audible warning system. The design of the truck shall incorporate the audible warning system specified in 3.11.8.1 through 3.11.8.4.

3.11.8.1 Sirens. The truck shall be equipped with a Federal Sign and Signal Corporation electronic siren and public address combination Model PA202A, or equal. The electronic siren shall be furnished with a microphone and all wiring applicable for connection to a two-way radio. A single, 6 inch dia speaker shall be mounted on a bracket between the sine-wave warning lights specified in 3.11.7.4.3. A Federal Sign and Signal Corporation siren Model Q2B, or equal . shall be mounted on the left side of the truck on a front bumper extension plate. Controls to activate the siren shall be two foot-operated buttons which are designed in conformance with MIL-STD-1472 requirements for foot operated switches. One button shall be located for driver left foot and one for passenger right foot use. An electric brake control for the siren shall be mounted on the dash in a location equally accessible by driver or passenger. The electric siren brake shall be designed so that it cannot brake a siren that is tarned on.

3.11.8.2 Horns. Two tone–adjustable, trumpet type, air activated horns shall be mounted on the cab roof outboard of the warning lights specified in 3. 11.11.5 or 3.11.12.10. The horns shall be powered with air from the brake air tank. The horns shall be controlled by a pull chain that is equally accessible by driver or passenger.

3.11.8.3 Buzzer. A buzzer shall be installed in the driver's compartment. The buzzer shall be controlled by a waterproof, push-button switch mounted on the rear of the truck, above the platform, in a position that permits platform passengers to signal the driver.

3.11.6.4 Back-up warning. An electrically operated back-up warning, in accordance with SAE J994, shall be installed to automatically operate when the truck transmission is shifted into reverse.

3.11.9 Handrails and grabrails. Handrails and grabrails shall be incorporated into the truck. The handrails and grabrails shall be 1.25 inch dia conforming with MIL–STD–1472 handle and grasp area requirements, and be heavy tubing mounted on brackets. The tubing shall be secured in the bracket to prevent rotation. As a minimum, the following handrails and grabrails shall be installed aboard the truck:

a. Two vertical grabrails, on either side of the truck body at the rear platform, to assist personnel mounting the platform.

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b. One horizontal handrail, not more than 66 inches above the platform. The handrail shall be located not more than 6 inches forward of the rearward end of the large-hose bin, or 6 inches forward of the clear area of the platform.

c. One horizontal. ribbed. rubber-covered stainless steel handrail located across the rear of the truck immediately below the end of the large-hose bin.

d. Two handrails. located atop the leader-hose bins, on the left and right sides of the truck. These handrails shall be the same length as the bins to which they are attached.

e. Two vertical grabrails shall be located on either side of the jump seat area to assist jump seat passengers up the steps into the jump seat area.

f. Two vertical grabrails shall be located on either side of the driver's compartment to assist driver and passenger entering the compartment.

g. Two horizontal handrails on the interior of each of the driver's compartment doors.

3.11.10 Stowed components. The stowed components to be furnished with the truck shall be soft suction hose, reducer, ladders, axes, pike poles, and wheel chocks. The holders, brackets or mounting pads used to mount the stowed components on the trucks shall be positive-locking, quick-release devices that keep the stowed components secure under all running conditions, yet permit quick removal.

3.11.10.1 Soft suction hose. One 12 ft length of 6 inch dia, commercial grade, soft suction hose shall be stowed in the soft suction hose compartment specified in 3.11.2.3. The hose shall be equipped with a 4.5 inch. long-handled , swiel type, female gasketed coupling on one end and a 6 inch, long-handled, swivel type, female gasketed coupling on the other end. The hose shall be connected to the front suction inlet. The hose coupling when connected shall not extend past the bumper.

3. 11.10.2 Reducer. One 6 inch NH to 4.5 inch NH reducer, with double female, double-swivel , long-handled, gasketed couplings shall be mounted on the inside of the through cabinet's streetside door.

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3.11.10.3 Ladders, Ladders shall be in accordance with NFPA Code No. 1931. The ladders furnished shall be as follows:

a. One 24 ft, two-section, extension type.

b. One 14 ft roof-type

c. One 10 ft attic-type

3.11.10.3.1 Ladder stowage. The ladders shall be stowed on the curbside of the truck. The 10 ft attic-type ladder shall be stowed on top of the curbside cabinets. The other two ladders shall be stowed vertically, on the side of the body, above the attic type ladder. The mounting height of the top edge of the two verticaliy mounted ladders shall be 51 inches, +5 inches. The two ladders shall be mounted together so as to avoid interference with the intake on the curbside of the truck. The ends of the two ladders shall have a padded cover to protect personnel from head injuries. Mounting devices shall be double-control-adjustment type that will prevent chafe and rattle.

3.11.10.4 Axes. Two axes shall be stowed on the jump seat area, on the exterior of the vertical cab extension. The axe mounted on the curbside shall be a pick-head fire axe, with a fiberglass handle. The axe mounted on the street-side shall be a flat-head, Dayton-pattern fire axe, with a fiberglass handle.

3.11.10,5 Pike poles. Three straight, plain-point and hook pike poles, with fiberglass handles, shall be stowed on the truck. One pike pole shall have a 10 ft long handle and another pike pole shall have a 6 ft long handle. These two pike poles shall be mounted in tubes atop the curbside cabinets, alongside the attic-type ladder. The third pike pole shall have a 32 inch long, “D” type handle, This shorter pike pole shall be mounted on the interior of the curbside door of the through cabinet.

3.11.10.6 Wheel chocks. Wheel chocks for each wheel of the truck shall be furnished and stowed in brackets beneath the running boards at the jump seat steps.

3.11.11 Type I truck. The type I truck shall be as specified herein and shall be equipped with a “booster hose system a deck pipe system and hard suction hose stowage, with hose.

3.11.11.1 Booster hose system. The booster hose system shall include a booster reel compartment, a booster hose reel, a booster hose and nozzle, piping and controls for the system, and hose rollers to protect the booster hose during play-out and operations.

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3.11.11.1.1 Booster reel compartment. A booster reel compartment shall be furnished at the rear of the truck, between the chassis frame members and ahead of the platform. the booster reel compartment shall have a roll–up type door and shall be equipped with drain holes in the floor.

3.11.11.1.2 Booster hose reel. A booster hose reel shall be installed in the booster reel compartment. The booster hose reel shall have a capacity of not less than 150 ft fo 1 inch rubber hose. The booster hose rell shall be an electric re-wind type, with a sealed pushbutton control on the reel, labeled in conformance with MIL-STD-1472 Labeling requirements, The reel shall include a provsion for handcrank re-wind. The booster hose reel shall be equipped with a friction brake to prevent overtravel during playout.

3.11.11.1.3 Booster hose and nozzle. Three 50 ft lengths of 1 inch inside dia, smooth bore, rubber-lined, chemical hose shall be installed on the booster hose reel. Each length of hose shall be fitted with reuseable, full-flow type couplings. The hose shall have a working pressure of not less than 600 psi. The nozzle shall be a 1 inch dia combination fog and straight stream type. The nozzle shall have a ball-shutoff type control that adjusts pattern and gallonage, Akron style 1710 or equal. The nozzle shall be attached to the hose and stowed in a quick-release holder inside the booster reel compartment. Mounting of the holder shall be as specified in 3.5.4.

3.11.11.1.4 Booster system piping and controls. An additional port shall be provided at the fire pump discharge manifold for the booster hose. The discharge line between the discharge manifold port and the discharge outlet in the booster reel compartment shall be provided with a discharge control valve as specified in 3. 11.5.3.3 and a discharge outlet drain as specified in 3.11.5.3.6. The discharge line, valve, drain and fittings shall withstand not less than 350 psi working pressure. Control for the booster discharge control valve shall be as specified in 3.11.6.3.1.6. Control for the drain shall be as specified in 3.11.6.3.1.9. The discharge outlet in the booster reel compartment shall be a 1 inch dia, NPT outlet.

3.11.11.1.7 Booster hose rollers. Booster- hose rollers shall be installed on the bottom of the booster reel compartment exit and on the rest-most vertical panels of the truck body. The vertical hose rollers shall be such that the booster hose shall be prevented from dropping below the rollers when the hose is played out toward the front of the truck. The hose rollers shall not extend beyond the rear platform and shall not substitute for any grabrail specified herein. The booster hose rollers shall be of corrosion-resistant steel that has been passivated in accordance with QQ-P- 35, or hot-dipped galvanized in accordance with ASTM A 153.

3.11.11.2 Deck pipe system. The deck pipe system shall include a deck pipe and the piping and controls for the system.

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3.11.11.2.1 Deck pipe. The deck pipe shall be an Akron style 506 or equal. with a discharge that shall be not less than 1000 gpm. The deck pipe wheel controls shall be capable of directing the pipe to any horizontal or vertical position, The horizontal control shall be furnished with a position lock. The deck pipe shall be equippped with a 1000 gpm, variable stream fog nozzle, Akron style 1750, or eqia. The deck pipe shall be mounted on the truck at the front, streetside corner of the large-hose bin.

3.11. 11.2.2 Deck pipe system piping and controls. An additionai port shall be provided at the fire pump discharge manifold for the deck pipe. The discharge line between the discharge manifold port and the deck pipe outlet atop the truck shall be provided with a discharge control valve as specified in 3. 11.5.3.3 and a discharge outlet drain as specified in 3.11.5.3.6. The discharge line, valve and fittings shall be sized to flow not less than 750 gpm. Control for the deck pipe discharge control valve shall be as specified in 3.11.6.3.1.6. Control for the drain shall be as specified in 3.11.6.3,1.9. A gage for the line pressure shall also be installed on the pump operator's panel and piped to the discharge line. The gage shall be as specified in 3.11.6.3.2.3.

3.11.11.3 Hard suction hose stowage. The hard suction hose storage facility" shall consist of suction hose trays or tubes and lengths of hard suction hose.

3.11.11.3.1 Suction hose trays or tubes. The truck shall be equipped with two suction hose trays or tubes. The suction hose trays or tubes shall be sized to stow 10 ft lengths of 6 inch dia hard suction hose with couplings. The trays or tubes shall be installed within the truck body. between the hose bins and the water tank top. The fasteners for the trays or tubes shall facilitate their removalfor access to the water tank top. The trays or tubes shall be equipped with positive-locking mechanical devices to prevent hose movement under all road conditions. The trays or tubes shall be equipped with drain holes over their entire length. The finish of the trays or tubes shall be such as to prevent hose chafe. The installation fasteners. hose locking devices, and the trays or tubes shall be of corrosion-resistant steel that has been passivated in accordance with QQ-P-35. Access to the trays or tubes shall be provided with doors as specified in 3.11.6.1.4.1, except that door locks shall not be furnished

3.11 11.3.2 Hard suction hose. One 10 ft length of 6 inch dia. commercial grade hard suction hose shall be stowed in each suction hose tray or tube. The hose shall have 6 inch dia NH, long-handled, female couplings on one and 6 inch dia NH, rocker-pin-lug, male couplings on the other end.

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3.11.11.4 Strainer One strainer, with a 6 inch dia NH female coupling, shall be stowed on the jump seat step riser, adjacent to the pump operator's panel.

3.11.11.5 Warning light, type I truck. A single warning light, containing not less than three sealed-beam rotating lamps, shall be centric-mounted on top of the truck cab. The warning light shall conform to NFPA Code No. 1901. The switch for this warning light shall be grouped with the switches specified in 3.11.7.5.2. The light shall have a red lens.

3.11.12 Type II truck, The type II truck shall be as specified herein and shall be equipped with a 50 ft long water tower. The water tower shall consist of a hydraulic system, turntable, boom and ladder combination tower, water tower water system, remote control nozzle, outriggers and a control station. The water tower shall be designed, constructed and powered to maintain stable performance and resist fracture or permanent deformation as follows:

a. With boom fullv extended, tip unsupported, flowing water up to 1000 gpm without men or load on the ladder, support and resist the reaction of the nozzle in any normal position of operation with the boom at any angle from 10° below horizontal up to 90° above horizontal.

b. With boom fully extended and raised to any angle between 45° and 75°, tip unsupported and flowing water up to 1000 gpm, support and resist the reaction of the nozzle in any position of operation plus a 300 lb load at ladder tip.

c. Used as a ladder only (no water flowing) with boom fully extended, tip unsupported and raised to any angle from 0° to 45°, support a tip load of 400 lbs.

d. Used as a ladder only (no water flowing) with boom fully extened, tip unsupported and raised to any angle between 45° and 75°, support a tip load of 800 lbs.

e. Used as a ladder only (no water flowing) with boom fully extended, tip suppore~d, carry and support a load of 1600 lbs evenly distributed along the length of the ladder.

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3.11.12.1 Hydraulic system, The hydraulic system used to actuate the water tower shall consist of all fluidpower components as specified herein. The fluid shall be oil conforming to MIL-L-2104, grade 10, or MIL-L-46167. All hydraulic system component materials shall be compatible with these oils. The hydraulic system shall have an oil capacity and a working pressure that is sufficient to operate the fluid power components Of the entire system to the extent required to operate the water tower as specified herein. The working pressure of the hydraulic system shall not exceed 3000 psi. System pressure shall be provided by an engine-driven pump that is powered by the truck engine. An interconnecting device between the hydraulic system and the truck engine throttle shall control truck engine speed as required by pressure and flow demands of the system. The hydraulic system shall incorporate a 12 Vdc, electric. motor-driven , back-up pump to provide a pressure source under emergency conditions.

3.11.12.1.1 Hydraulic fittings. The hydraulic fittings on pumps. valves and cylinders shall be either O-ring, straight thread type as specified in SAE J514, or the 4-bolt flange type as specified in SAE J518 except only 4-bolt flange type fittings as specified in SAE J518 shall be used for ports larger than 1 inch.

3. 11.12.1.2 Hydraulic lines. Hydraulic lines shall consist of tubing, flexible hose and steel fittings. Hose or combinations of hose and tubing shall be used between components Chat are not mounted to a nonflexing, rigidly connected structural member or rigidly connected subassembly. All lines shall be routed to provide a minimum number of bends, ease of maintenance, maximum protection and to avoid chafing. The hose installation, such as angle of fittings and location shall cause no stress concentration on the hose at tune fiitting, and hose shall be supported as necessary to eliminate sagging,at the fittings. All lines, other than suction lines or cylinder bleeder lines, shall be considered pressure lines.

3.11.12.1.2.1 Pressure hose assemblies. Pressure hose shall be as specified in SAE J517 or MIL-H-52471. Permanently attached type pressure hose couplings are acceptable on original equipment hose assemblies. The pressure hose inside dia shall be limited to 1/4, 3/8, 1/2, 3/4, 1, 1-1/4, 1-1/2 and 2 inches. The hose used shall have a rated operating pressure that is equal to or greater than the hydraulic circuit maximum relief valve setting. Hose installation shall not cause the hose to be bent beyond the minimum bend radius specified in SAE J517.

3.11.12.1.2.2 Pressure tubing. Pressure tubing shall be in accordance with SAE J524 or SAE J525. The pressure tube sizes shall be limited to 1/4, 5/16, 3/8, 1/2, 5/8. 3/4, 1, 1-1/4, 1-1/2 and 2 inches. Tubing shall have a burst pressre that is not less than 3.0 times the system maximum working pressure. All tubing shall be securely clamped and all bends shall be smooth without flattening (see 6.4.4), kinking or wrinkling of the tube.

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3.11.12.1.2.3 Pressure tube fittings. Pressure tube fittings shall conform to SAE J514. Jump size tees may be used. Pressure tube fitting sizes shall be limited to the sizes specified for pressure tubing. Pressure tube fittings shall have a burst pressure that is not less than 3.0 times the system maximum working pressure.

3.11.12.1.2.4 Suction lines. All pump inlet suction lines shall incorporate a flexible means of absorbing vibration or movement between the pump and reservoir. The suction line shall be sized so that the pump inlet pressure does not drop below manufacturer's rating at 140°F, + 10°F oil temperature and at governed engine speed. The suction hose shall conform to SAE J517, 100R4. Suction hose clamps shall be of the double wrap stainless steel type in accordance with SAE J536, type F, style 4.

3.11.12.1.1.5 Return lines. Return lines shall be in accordance with SAE J517.

3.11.12.1.3 Hydraulic pump. The pump manufacturer's 1000 hour pump contaminant tolerance profile of the pump(s), shall not fall below or cross the $^{B}10$ curve of the hydraulic system filters, when the filters have been tested in accordance with MIL-F-52723; or, the pump Omega rating shall be less than the filter Beta 10 rating.

3.11.12.1.4 Hydraulic system filtration. Hydraulic system filters shall be furnished in the pressure line, return line, or both.

3.11.12.1.4.1 Return line filters. The system filter element shall be in accordance with the performance requirements of MIL-F-52723. The element need not be qualified under MIL-F-52723. The filter element shall maintain a full flow condition for the duration, of all testing specified herein. The filter bypass may open for system oil at a system temperature that is less than 70°F. All return oil shall pass through the filter, except relief oil which may return directly to the reservoir. The filter shell be located so that the element may be removed for examination and replacement without spilling or draining oil from the reservoir. At maximum system oil flow through the filter, including boom depression at maximum governed engine speed and an oil temperature of 150°F, + 5°F, the pressure drop across a new filter assembly shall not exceed 10 psi. An automatic bypass shall be provided in the system filter and shall open at not less than 10 psi more than the pressure drop measured across the new filter assembly and shall permit full flow at a prressure drop of not more than 35 psi. The filter housing shall have a proof pressure of not less than 100 psi without leakage, damage or permanent deformation. The filter assembly shall be equipped with provisions for a remotely mounted contamination indicator which will indicate when the filter is clean and when the filter is bypassing.

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3.11.12.1.4.2 Pressure line filter. The pressure line filter shall be as specified in 3.11.12.1.4.1, except that the proof pressure of the filter housing shall be not less than 150 perent of the maximum system operating pressure.

3.11.12.1.5 Hydraulic fluid reservoir. A hydraulic reservoir shall be furnished to contain hydraulic system oil. The reservoir shall be mounted to resist vibrations that produce leaks, loosening of fittings or fasteners, brinelling of material, or cracks. The reservoir shall have sufficient capacity to prevent air from enterning the hydraulic lines when all pistons are fully extended and sufficient free air capacity to prevent oil from being discharged through the reservoir air vent when maximum return oil is being surged into the reservoir. A sightglass or dipstick, with appropriate "Add" and "Full" marks shall indicate the correct oil levels when the truck is standing on level ground and the water tower is in the riding position. A filler opening shall be located to provide for filling of the reservoir from a 5 gallon container conforming to MIL-C-1283 while standing on the ground or a riding surface of the truck. The neck of the filler opening shall contain a removable 50 mesh screen, or finer, strainer. The filler tube shall be not less than 1 inch inside dia. The filler opening shall be sealed with a removable filler cap attached to the truck with a durable chain. The suction line connections shall be not less than 2 inches above the reservoir's bottom. Return line connections shall discharge into the reservoir below oil level. The suction and return line connections shall be located on different sides of the reservoir, anti interior baffling inside the reservoir shall provide a tortuous flow path that will prevent a direct current between ports that connect the pump and parts that connect return lines. The reservoir shall be equipped with one or more access holes, with covers, for inspection and cleaning.

3.11.12.1.6 Pressure relief valve. A pressure relief valve shall be provided to limit pump pressure. Pressure equal to the relief valve setting shall be generated only when a cylinder is bottomed. The relief valve setting shall not exceed the pump manufacturer's rating.

3.11.12.1.7 Cylinders. All hydraulic cylinders shall be fabricated of seamless steel tube, or welded and redrawn tube. Ail welding and brazing of the cylinder- within the zone of piston operation shall be done before final machine honing or rolling of the cylinder. All internal cylinder ports shall be located beyond the area of piston travel. Each cylinder shall be equipped with an exclusion device that prevents foreign material and fluids from entering around the piston rod and damaging seals and other cylinder components. Each piston shall have a bearing surface concentric with the piston to carry piston side loads against the cylinder wall. The bearing surface shall be of a material that will not scar, score or gall the inside of the cylinder. Cylinder rods shall be hard chrome plated. All cylinders shall perform without buckling or bending and shall withstand a pressure that is equal to 1.5 times the relief valve setting without. evidence of rupture, permanent deformation, internal or external leakage or any form of visible damage. Cylinders shall be located for ease of maintenance and replacement.

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3.11.12.1.8 Hydraulic system cleanliness. The particle counts in the hydraulic system after installation of all fluidpower components, and flushing, and cleaning of only system before delivery, shall not exceed 1000 particles per milliliter greater than 10 micrometers and 20 particles per milliliters greater than 20 micrometers.

3.11.12.1.9 Hydraulic leakage. After completion of the pumping stability and ladder stability. tests (see 4.5,5.1 and 4.5.5 .2), the hydraulic system leakage shall not excced class 4, for dust-free conditions, 2nd class 4D, for dusty conditions, of SAE J1176.

3.11.12.2 Turntable. The turntable of the water tower shall be rotated by a hydraulic motor driving through a ring and pinion gear set. The turntable shall be capable of rotating in continuous 360° turns in either direction. The turntable top shall be not less than 36 inches in dia and shall have raised pivots for mounting the boom and ladder combination which will permit boom depression below the horizontal. Control of the turntable rotation shall be incorporated into the one-handed water tower hydraulic control (see 3.11.12.7.1) mounted on the panel of the control station (see 3.11.12.7).

3.11.12.3 Boom and ladder combination. The boom and ladder combination shall be a telescoping section manufactured of aluminum boxbeam. The boom and ladder combination shall be provided with a recess on the underside of the outer end to accommodate a fully depressed remote control nozzle (see 3.11.12.5), with the nozzle tip in a retracted position. The recess shall be sized to completely protect the depressed remote control nozzle without the use of external guard rails. When fully extended to maximum operating height, the boom and ladder tip shall be not less than 49 it above level ground. The horizontal reach of the boom and ladder in the fully extended condition at 0° over the side of the truck shall be not less than 40 ft. All measurements of the boom and ladder combination shall be in accordance with NFPA Code No. 1901. The boom and ladder combination shall be hydraulically actuated and shall be capable of extension, retraction, elevation and depression while the water tower water system (see 3.11.12.4) is flowing water at the rate of 1000 gpm through the remote control nozzle. Control of the boom and ladder combination shall be incorporated into the one-handed water tower hydraulic control mounted on the control station panel.

3 .11.12.4 Water tower water system. The water tower water system shall be routed through a 4 inch dia master control valve. Water shall be supplied to the master control valve by piping from the fire pump discharge manifold and by piping from two 2.5 inch dia inlets mounted on the rear of the truck. The piping between the master control valve and the rear-mounted inlets shall be equipped with properly-sized, balanced , corrosion–resistant full-flow-through type valves designed for remote control. Piping and connections between the master control valve and the remote control nozzle shall be capable of providing a discharge rate of 1000 gpm of water when:

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a. The turntable is rotated through any horizontal are up to 360°.

b. The boom and ladder combination is either fully extended or fully retracted.

c. The boom and ladder is completely horizontal or at a vertical height of not less than 49 ft.

d. The truck is sitting on a level, hard surface and the outriggers (see 3.11.12.6) are in the operating position.

3.11.12.4.1 Master control valve. The master control valve shall be designed for remote or through-panel control. The valve shall be of corrosion-resistant metal. The control for the master control valve shall be mounted on the control station panel.

3.11.12.4.2 Water tower inlets. The rear mounted water tower inlets shall be furnished with a rocker-lug, male plug and attachment chain for each inlet.

3.11.12.4.3 Inlet drains. The piping between the inlets and the master control valve shall incorporate drains as specified in 3.11.5.3.6, except that controls shall be as specified in 3.11.12.7.

3.11.12.5 Remote control nozzle. A remote control nozzle (monitor) shall be mounted on the outer end of the boom and ladder combination of the tower. The monitor shall be equipped with a remotely controlled nozzle tip (stream shaper) that will provide a uniform stream of water at any flow rate between 300 gpm and 1000 gpm. The monitor shall be capable of being elevated or depressed in a vertical plane of not less than 250°. The monitor shall be capable of being swept laterally through a 90° arc (45° to either side of center). The functions of elevation and depression in the vertical plane shall be completely independent of the sweep function in the horizontal plane. The functions of nozzle elevation, depression and sweep shall be hydraulically actuated. A set of controls for the monitor shall be mounted near the monitor which can be operated from the top of the ladder and another set of controls shall be mounted on the control station panel.

3.11.12.6 Outriggers. The boom and ladder combination shall be equipped with hydraulically actuated outriggers. The outriggers shall be arranged to stabilize the truck when the boom and ladder combination is raised and flowing water. The outriggers shall be self-locking and equipped with flashing red lights to warn of outrigger extension. The red warning lights shall activate automatically when the outriggers are extended. Controls for the outriggers shall be mounted on the water tower control station.

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3.11.12.7 Control station. A control station shall be provided on the rear of the truck. The control station panel shall be sized to accommodate the controls. gages and switches specified in 3. 11.12.7.1 through 3.11.12.7.7. The controls,. gages and switches on the control station panel shall be in accordance with the Controls requirement of MIL-STD-1472. Labels and instructions for the controls, gages and switches on the control station panel shall be in accordance with the Labeling requirements of MIL–STD-1472, and be provided with switch controlled illumination of not less than 30 cd/m^2.

3.11.12.7. 1 Water tower hydraulic control. A water tower hydraulic control that combines turntable rotation, boom elevation and depression, and boom extension and retraction by a one-handed operation shall be mounted on the right Side of the control station panel.

3.11.12.7.2 Water system control. A control handle for the 4 inch dia master control valve shall be mounted on the control station panel beneath the gage specified in 3.11.12.7.3. The control shall be rotary.

3.11.12.7. 3 Water pressure gage. A water pressure gage shall be mounted on the control station panel. The water pressure gage shall have a dial pressure range of 30 inches of Hg to a pressure of not less than 600 psi. The dial face of the gage shall be not less than 3.5 inches dia. The gage shall be installed on the panel with a plate or collar, and shall be piped to the inlet side of the master control valve.

3.11.12.7.4 Remote nozzle controls. Separate controls for the remote control nozzle shall be mounted on the control station panel in near proximity to each other. The controls shall be:

a. Nozzle vertical control to elevate and depress the nozzle in a vertical plane.

b. Nozzle horizontal control to sweep the nozzle back and forth.

c. Stream shaper control to adjust the nozzle tip.

3. 11.12.7.5 Outrigger controls. A control to extend and retract the outriggers shall be mounted on the control station panel. The control shall be provided with an automatic means to render the outriggers inoperative (lockout) when the boom and ladder combination is elevated. A protected switch that unlocks the self-locking feature of the outriggers shall be mounted in near proximity to the outrigger control on the panel. Protection for the switch shall completely surround the switch in such a manner that only intentional switching can be accomplished.

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3 11.12.7.6 Master hydraulic electric switch. A master hydraulic electric switch that controls the electric power circuit to the 12 Vdc electric-motor-driven pump shall be mounted on the control station panel. The master hydraulic electric switch shall be equipped with a red-bezeled indicating light that will activate whenever the master hydraulic electric switch is in the "on" position.

3.11.12.7.7 Throttle control switch. A throttle control switch shall be mounted on the control station panel and labled in accordance with MIL-STD-1072 requirements for labels. Activation of this switch by the operator shall prevent engine surging when minor hydralic control movements are initiated. The throttle control switch shall be a spring-loaded, normally-closed type of switch.

3.11.12.8 Communication system. A communication system designed in conformance with MIL-STD-1472 speech transmission and reception requirements, between the control station and the top of the boom and ladder combination shall be provided. The communication system shall allow clear communication between personnel at the control station and personnel at the remote control nozzle controls at the top of the boom and ladder combination.

3.11.12.9 Pull-out threadplate. A pull–out treadplate shall be provided beneath the running board at the pump operator’s panel. The threadplate shall be capable of supporting not less than 300 lbs on its outermost edge. The width of the treadplate shall be not less than 80 percent of the width of the pump operator 's panel and the front-to-hack depth shall be not less than the running board width. A positive-. locking mechanism shall be provided to secure the treadplate in the closed (retracted) position.

3.11.12.10 Warning light, type II truck. Two warning lights, containing not less than three sealed-beam rotating lamps, shall be centric-mounted on top of the truck cab on either side of the boom and ladder riding position. The warning lights shall conform to NFPA Code No. 1901. A single switch shall control both lights and shall be grouped with the switches specified in 3.11.7.5.2. The lights shall have red lenses.

3.12 Physical protection.

3.12.1 Lubrications. The truck shall be serviced with military lubricants selected by the contractor in accordance with table I, SAE J754. Run-in and testing shall be performed with these lubricants as applicable for the ambient operating temperature. A lubrication data plate or commercially proven decals in accordance with SAE J753 shall be provided and mounted (see 3.8.2.2) on each truck.

3.12.1.1 Lubrication fittings. Lubrication fittings shall conform, to SAE J534. Fittings shall be located in a protected, accessible position. Fittings shall be accessible to a grease gun conforming to MIL-G-3859, equipped with a hydraulic coupler conforming to MS24207 and a 10 inch flexible extension. lubrication fittings shall be provided without removal or adjustment of accessories or parts. Panels or plates equipped with hand-operated fasteners may be removed to provide accessibility. Accessibility shall be in accordance with SAE J925 for personnel normally clothed.

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3.12.1.2 Enclosures with integral reservoir of lubricant. Enclosures such as engine crankcase, transmission housing, and gearcases which contain a reservoir of liquid lubricant shall be equipped with dipsticks marked in accordance with SAE J614, check plugs not less than 0.5 inch pipe size, or sight gages to determine the acceptance level of the lubricant. Each enclosure shall be equipped with a means for filling the enclosure with lubricant and for draining the lubricant. Drains shall be fitted with an SAE J531 or J532 magnetic drainplug. The drainplug shall be located so that removal of the plug will result in complete drainage of the lubricant. Drainage shall be to a container on the ground when the truck is on level ground. Integral tubes or troughs may be used to convey the lubricant from the drain to the container. Accessibility to the drainplug, the filling means, and the lubricant-level checking device shall be provided without removal or adjustment of accessories or parts. Panels or plates equipped with hand-operated fasteners may be removed to provide accessibility. Accessibility shall be in accordance with SAE J925 for personnel normally clothed.

3.12.2 Fungus and moisture resistane. Electrical circuitry, including all components and connections, except as specified below, shall be protected from the effect of fungus growth and moisture penetration by an overall treatment with a varnish conforming to MIL-V-173, composition I:

a. Components or circuit elements which are inherently fungus and moisture resistant. or which are hermetically sealed need not be treated.

b. Components or circuit elements whose function will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to provide a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that surfaces are free from all foreign matter that would interfere with the adherence or function of the varnish.

3.12.2.1 Varnish composition. When local air pollution regulations preclude the use of MIL-V-173, composition I varnish, composition II varnish may be used for fungus and moisture resistance. When composition II varnish is used, the contractor shall have evidence on hand that the use of composition I varnish is precluded and that the composition II varnish complies with Rule 442 of the South Coast Air Quality Management District.

3.12.3 Rustproofing. When specified (see 6.2) , the truck shall be rust-proofed in accordance with MIL-STD-1223.

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3.13 Electromagnetic interference (EMI). Unless otherwise specified (see 6.2), the EMI emission and susceptibility characteristics of the truck shall be in accordance with SAE J551.

3.14 Dimensions. The overall dimensions of the truck, including all lights and mounted equipment, except mirrors, shall not exceed the dimensions specified in table I.

TABLE I. Truck dimensions.

Truck type	Height (inches)	Length (inches)	Width (inches)
Type I	115.0	336.0	96.0
Type II	132.0	384.0	96.0

3.15 Weight. The gvw of the truck, when loaded as specified in 3.7.1, shall be not more than the manufacturer's gvw ratings for the axles and tires.

3.16 Color. Unless otherwise specified (see 6.2), those parts of the truck which arc normally painted, shall be gloss lime-yellow. color No. 13670 of FED-STD-595.

3.17 Finish. The truck exterior shall be cleaned, treated and painted in accordance with MIL-T-704, type F for ferrous metals and type G for non-ferrous metals. Parts and components may be primed with alkyd base primer which has been allowed to cure for a minimum of 14 days and then cleaned and painted as specified herein either prior to, or at, final truck assembly painting. Treatment and painting of the chassis frame. undercarriage and underbody shall be in accordance with MIL-STD-1223.

3.18 Identification marking. The truck shall be identified in accordance with MIL-STD-130. Identification name plates and parts data plates or labels shall be in accordance with MIL-STD–1223. The plate shall conform to MIL-P-514, type I, style 1. composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets.

3.18.1 Exterior markings. When specified (see 6.2) , exterior markings on the truck shall be in accordance with MIL-STD-1223.

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3.19 Workmanship.

3.19.1 Castings and forgings. All parts, components, and assemblies of the truck which include castings and forgings shall be free of harmful extraneous material such as sand, dirt, sprues. scale and flux, Rework shall be limited to procedures which do not reduce mechanical properties or affect function.

3.19.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing. All bends shall be made with controlled means to insure uniformity of size and shape. precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free from burrs, sharp edges and corners, excep when sharp edges or corners are required or where they are not detromental to safety.

3.19.3 Bolted connections. Boltholes shall be accurately formed and shall have burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or cap screws shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement. Bolts shall protrude not more tnan two full threads.

3.19.4 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member, and shall be in accordance with SAE J492, or Industrial Fastener Institute Standard No. 114 for break mandrel blind rivets.

3.19.5 Welders and welding.

3.19.5.1 Welders. Before assigning any wleder to manual welding work covered by this specificaation, the contractor shall provide the contracting officer with certification (see 6.2.2)that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

ANS/ASME	Boiler and Pressure Vessel Code, Section IX. Welding Qualifications.
ANSI/AWS D1.1	Structural Welding Code, Section 5, Qualification.

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Contractors who make only horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor' s procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with ANSI/ASME or ANS/ASME codes.

3.19.5.2 Welding. The surface of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping, and wire brushing. and other foreign matter. Parent materials, weld filler metals and fabrication techniques shall be as required to enable the truck to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separared by more than 3/16-inch unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the truck shall be at the option of the contractor.

3 .19.5.3 Aluminum welding. The aluminum surfaces of parts to be welded shall be brushed with a stainless steel shrl:sh brush until free from scale paint, grease. oxide film, and other foreign matter. The welding shall be done by the inert gas, shielded arc method. Welds may be ground, filed, buffed, or chipped but shall not be hammered and shall be free from discontinuities such as base metal cracks adjacent to or behind welds, cracks in weld metal, undercutting of base metal, lack of fusion with parent metal or between parts, lack Of penetration, lack of prescribed fit, or splatter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibililty for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of ail inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government , The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and service conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

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a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 First article examination. The first article shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection of the first article.

4.3.2 First article tests. The first article shall be subjected to the tests marked "X" in column 1 of table II in the order indicated in the table. Failure of any test shall be cause for rejection of the first article.

Table II. Test schedule.

First article	Quality conformance	Test	Requirement paragraph	Test paragraph
1	2	3	4	5
X	X	Hydraulic system cleanliness (ty II)	3.11.12.1.8	4.5.5.3.6
X	X	Hydraulic filters (ty II)	3.11.12.1.4.1, 3.11.12.1.4.2	4.5.5.3.4
X	X	Hydraulic cylinder proof (ty II)	3.11.12.1.7	4.5.5.3.5
X	X	Hydraulic pump contaminent tolerance (ty II)	3.11.12.1.3	4.5.5.3.3
X		Hosebin and cabinet bottoms	3.11.6.1.1.1, 3.11.6.1.2, 3.11.6.1.3, 3.11.6.1.4, 3.11.6.2.1	4.5.2.12
X		Treadplate load (ty II)	3.11.12.9	4.5.5.6
X		Exterior noise	3.9.1	4.5.2.4
X		Personnel noise	3.9.2	4.5.2.4
X		Alternator output	3.11.1.7	4.5.2.8.1
X		Inverter voltage control	3.11.2.1.7	4.5.2.8.2

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Table II. Test schedule. - Continued

First article	Quality conformance	Test	Requirement paragraph	Test paragraph
1	2	3	4	5
X		EMI	3.13	4.5.3
X	X	Mobility	3.7.1	4.5.2.2
X	X	Service brakes	3.11.1.6.1	4.5.2.7.1
X		Parking brakes	3.11.1.6.2	4.5.2.7.2
X	X	Engine brake	3.11.1.6.3	4.5.2.7.3
X	X	Pumping capacity	3.7.2	4.5.2.3
X		Fuel system capacity	3.11.1.2.3	4.5.2.6
X		Pump drive temperature	3.11.5.1.1	4.5.2.3.1
X		Discharge relief valve	3.11.5.3.2	4.5.2.3.2
X		Water tank pumping	3.11.3.4	4.5.2.9
X		Foam system	3.11.4	4.5.2.10
X	X	Deck pipe line flow (ty I)	3.11.11.2.2	4.5.4.1
X		Pumping stability (ty II)	3.11.12a. and 3.11.12b.	4.5.5.1
X		Altitude pumping (see 6.2)	3.10.1	4.5.2.5.1
X	X	Hydrostatic	3.11.5	4.5.2.11
X		Ladder stability (ty II)	3.11.12c., 3.11.12d., 3.11.12e.	4.5.5.2
X		Throttle control switch (ty II)	3.11.12.7.7	4.5.5.3.9
X		Turntable rotation (ty II)	3.11.12.2	4.5.5.3.8
X		Hydraulic back-up (ty II)	3.11.12.1	4.5.5.3.1
X		Hydraulic pump inlet pressure (ty II)	3.11.12.1.2.4	4.5.5.3.2
X		Hydraulic external leakage (ty II)	3.11.12.1.9	4.5.5.3.7
X		Cold starting and operating	3.10.2	4.5.2.5.2
X		Hot starting	3.10.3	4.5.2.5.3
X		Maintainability	3.6	4.5.2.1

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4.4 Quality conformance inspection.

4.4.1 Examination. Each truck furnished shall be examined for the defects specified in 4.5.1. Presence of any critical defect, five or more major defects or two or more minor defects shall be cause for rejection of the truck.

4.4.2 Tests. Each truck shall be subjected to the tests marked "X" in column 2 of table II, Failure of any test shall be cause for rejection of the truck.

4.5 Inspection procedure.

4.5.1 Examination. The truck shall be examined as specified herein for the following defects:

Critical

Defect No.	Defect	Requirement Paragraph
1.	Master pressure gage not compound type or with dial face size less than specified.	3. 11.6.3.2.1
2.	Master suction gage not compound type or with dial face size less than specified.	3. 11.6.3,2.2
3.	Back-upp warning not as specified.	3.11.8.4
4.	Truck's gvw exceeds gvw ratings for axles and tires.	3.15

Major

Defect No.	Defect	Requirement Paragraph
101,	Materials not as specified.	3.3
102.	Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.	3.3.1
103.	Dissimilar metals as defined in MIL-STD–889 are not effectively insulated from each other.	3.3,1.1
104.	Contractor does not have documentation available for identification of material, material finishes, or treatment,	3.3.1.2
105.	Used, rebuilt or remanufactured components. pieces, or parts incorporated in the truck.	3.3.2
106.	Design of the truck not as specified	3.4
107.	Angles of approach and departure less than specified in NFPA Code No. 1901.	3.4.1
108.	Construction of the truck not as specified.	3.5

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Defect No.	Defect	Requirement Paragraph
109.	Interchangeability of parts not as specified.	3.5.1
110.	Threads of the piping, not in conformance with FED-STD-H28.	3.5.2
111.	Wiring not identified; identification not consistant; protection. installation, ant! attachment not as specified; or. fuses used or circuit breakers omitted.	3.5.3
112.	Mounting not secured to the truck as specified.	3.5.4
113.	Attaching chains not in conformance to RR-C-271, type II, clcss 6, corrosion resistant. or zinc coated or not attached as specified.	3.5.5
114.	major components not removable and replaceable or access openings not sized, finished and covered as specified.	3.6
115.	Operator characteristics not as specified.	3.8.1
116.	Control labels not in accordance with MIL-STD-1472 or universal symbology for driver's gages, in-struments and controls rot in accordance with SAE J1500	3.8.2.1
117.	Identification and instruction plates not installed or not of material specified.	3.8.2.2
118.	Personnel protection not as specified or protective devices impair operating efficiency of the truck.	3.9.3
119.	Boarding and riding protection not as specified.	3.9.4
120.	Chassis not 4x2, from-wheel-steer, rear-wheel-drive type.	3.11.1
121.	Frame not heavy-duty fire service type or altered so as to reduce its strength.	3.11.1.1
122.	Engine selection, accessories or type net as specified.	3.11.1.2
123.	Engine not 280 bhp at 2000 rpm, or engine not EPA certified.	3. 11.1.2.1
124.	Contractor does not have bhp certification available.	3.11.1.2.1.1
125.	Engine lubrication system not as specified.	3.11.1.2.2
126.	Engine fuel system filler opening, cap position, fuel line installation or tank position not as specified.	3.11.1.2.3
127.	Engine cooling system not of capacity recommended or not of materials specified.	3.11.1.2.4
128.	Engine starting system allows engine start with transmission in other than neutral position or ignition switch is not keyless with a pilot light.	3.11.1.2.5

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Defect No.	Defect	Requirement Paragraph
129.	Engine exhaust system does not include components specified or is not installed as specified	3.11.1.2.6
130.	Any component of drive assembly with a torque capacity rating less than the maximum torque developed by the engine.	3.11.1.3
131.	Transmission not fill automatic, 4–speed with reverse shift protection and neutral locking device.	3.11.1.3.1
132.	Drive axle not full floating, type.	3.11.1.3.2
133.	Springs not of rated capacity,	3.11.1.4.1
134.	Shock absorbers not double-acting, hydraulic type or not matched to suspension capacity.	3.11.1.4.2
135.	Wheels and tires do not conform to TRA requirements for truck GVW at 50 to 60 mph.	3.11.1.5
136.	Front and rear wheels not 10 hole disk type, not interchangeable or not inherently in balance.	3.11.1.5.1, 3.11.1.5.2
137.	Tires not at least 100 level quality, not sized for	3.11.1.5.3
	Tire valves, caps and cores not as specified.	
138.	Tire valves, caps and cores not as specified.	3.11.1.5.3.1
139.	Spare wheel and tire not furnished or not secured in hose bin.	3.11.1.5.4
140.	Brakes do not conform to latest DOT requirements.	3.11.1.6
141.	Service brake mechanisms not accessible for adjustment, or service brake lines not installed and protected as specified.	3.11,1.6.1
142.	Parking brake not all wheel locking, not type specified, or without emergency release mechanism.	3.11.1.6.2
143.	Engine brake not of type specified.	3.11.1.6.3
144.	Generating system has alternator of less than 145 amp output at 12 Vdc.	3.11.1.7
145.	Steering system not hydraulic power–assisted, or provides a cramping angle of less than 28°.	3.11.1.8
146.	Rear platform not constructed to serve as bumper or with riding surface more than 25.5 inches above level ground.	3.11.1.9.1
147.	Front bumper not heavy duty unit attached to frame.	3.11.1.9.2
148.	Towing hook or towing lugs do not permitt attachment of a medium duty towing bar, or reduce the angles of approach and departure.	3.11.1.10
149.	Towing hook does not provide 6 inch clearance below platform.	3.11.1.10.1
150.	Towing lugs do not conform to MS500004, 10 to 50 ton.	3.11 .1.10.2

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Defect No.	Defect	Requirement Paragraph
151.	Partition in cab does not have sliding glass window or window is less than 12 inches wide by 6 inches high.	3.11.2
152.	Driver's compartment not separated from engine by insulation,on firewall or not finished. trimmed and sealed as specified.	3 .11.2.1
153.	Driver's compartment seats not upholstered as specified, or undersized; driver's seat not adjustable.	3,11 .2.1.1
154.	Driver's dashboard does not extend full width of cab, lacks specified depth, or permits dash lighting to reflect on windshield.	3. 11.2.1.2
155.	Driver's gages and instruments not installed as specified, are fewer than minimum required, or are not lighted by headlight switch control.	3.11.2.1.2.1
156.	Driver's operating controls not installed as specified.	3,11 ,2.1.2.2
157.	Gage and instrument monitors not as specified.	3.11.2.1.2.3
158.	Glass and glazing not of quality or not sealed as specified.	3.11.2.1.3
159.	Windshield wiping and cleaning system not in conformance with SAE J198 or does not have reservoir of at least 2 quart capacity.	3.11.2.1.4
160.	Heater in driver's compartment not as specified.	3.11.2.1.5
161.	Transmission shift control pedestal quadrant or location not as specified.	3.11.2.1.6
162.	Inverter not equipped with switch and voltmeter, or not located as specified.	
163.	Communications compartment omitted, not of size specified, or not adapted to communications equipment.	3.11.2.1.8
164.	Rear-view mirrors not correct type, quantity or mounting; or stick-on convex mirror used.	3.11.2.1.9.1
165.	Seat belts and anchors not as specified. or without correct buckles and retractors.	3. 11.2.1.9.2
166.	Interior lights not correctly located or not correctly switched.	3.11 .21.9.3
167.	Dual sun visors not installed in driver's compartment.	
168.	Jump seat area not separated by engine cover.	3.11.2.2
169.	Engine cover not ventilated or ventilating louvers direct engine compartment air toward jump seat occupants.	3.11.2.2.1

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Defect No.	Defect	Requirement Paragraph
170.	Jump seats not sized, upholstered or bracketed as specified.	3 .11.2.2.2
171.	Jump seat safety bars not size, type or locked as specified.	3 .11,2.2.2.1
172.	Jump seat dome lighy omitted or without control switch.	3.11.2.2.2.2
173.	Soft suction hose compartment undersized, with projections inside, without drain holes, or not finished to avoid chafing hose.	3.11.2.3
174.	Water tank attached to any part of truck body.	3 .11.3
175.	Water tank size. shape, baffling, top, or top fastening not as specified.	3 .11.3.1
176	Water tank not manufactured of correct materials or not cleaned as specified be before installation.	3 .11,3.2
177.	Water taank fill opening less than 64 square inches, unsecured, without specified closure, or overflow drain provisions as specified.	3.11.3.3
178.	Foam system not of type or capacity specified; or not of corrosion-resistant materials.	3.11.4
179.	Foam liquid tank not removeable or not located above fire pump.	3.11.4.1
180.	Foam liquid tank not of 60 gallon capacity or not baffled to prevent excessive surging.	3.11.4.1.1
181.	Foam liquid Lank not manufactured of correct materials or not cleaned as specified before installation.	3.11.4.1.2
182.	Foam liquid tank filler opening not sized, screened or covered as specified.	3.11.4.1.3
183.	Foam liquid tank not vented, or with vent piping not as specified.	3.11.4.1.4
184.	Foam liquid Lank outlet not screened.	3.11.4.1.5
185.	Foam liquid tank not equipped with drain, drain valve , or drain pipe directed as specified.	3.11.4.1.6
186.	Foam liquid metering valve not as specified.	3.11.4.2
187.	Foam system valving and piping not as specified.	3.11.4.3
188.	Foam system flushing provisions omitted or do not provide for complete flushing of system.	3.11.4.3.1
189.	Foam system controls not as specified or without instruction plate.	3.11.4.4
190.	Fire pump nor UL rated or priming device does not conform to UL 822.	3.11.5.1
191.	Suction section not of corrosion-resistant metal or not galvanized as specified.	3.11.5.2
192.	Fire pump suction manifold not as specified.	3.11.5.2.1

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Defect No.	Defect	Requirement Paragraph
193.	Large suction inlets not of size or location; without strainers; or not valved or capped, as specified.	3.11.5.2.2
194.	Small suction inlets net of size, type or location specified or not plugged as specified.	3.11.5.2.3
195.	Water tank suction piping without shut-off and check valves .	3.11.5.2.4
196.	Suction manifold fittings at ports not corrosion-resistant.	3.11.5.2.5
197.	Discharge section not of corrosion-resistant metal or riot galvanized as specified.	3.11.5.3
198.	Fire pump discharge manifold not as specified.	3.11.5.3.1
199.	Indicator light provision not incorporated into pressure relief valve.	3.11.5.2.2
10la.	Discharge control valves not of type specified, or not numbered and tagged as specified.	3.11.5.3.3
102a.	Discharge outlets not of quantity, size or location, or not labelled as specified.	3.11.5.3.4
103a.	Discharge piping not as specified.	3.11.5.3.5
104a.	Discharge outlet drains of wrong size, improperly located, or with drain lines not as specified.	3.11.5.3.6
105a.	Truck body does riot provide for attachment of storage units or is not reinforced at points of attachment	3.11.6
l06a.	Hose bins and through cabinet not integral with the truck body.	3.11.6.1
l07a.	Small hose bin not located, sized, divided, and with smooth interior, or without removable bottom and floor, or without hose rollers as specified.	3.11.6.1.1
108a.	Any hose bin bottoms not removeable; any hose bin and cabinet bottoms without raised crossmembers with attachments for floors, or without drain holes between the crossmembers.	3.11.6.1.1.1. 3.11.6.1.2, 3.11.6.1.3, 3.11.6.1.4, 3.11.6.2.1
109a.	Any hose bin and cabinet floors not of materials specified, with slat sizes or slat spacing not as specified , or not fastened to bottom crossmembers by counter-sunk screws or bolts.	3.11.6.1.1.2
110a.	Large nose bin not located, sized, divided. and with smooth interior, or without removeable bottom and floor.	3.11.6.1.2
111a.	Fitted cover for large hose bin not provided, or fastening methods not as specified; design and material not as specified.	3.11.6.1.2.1

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Defect No.	Defect	Requirement Paragraph
112a.	Leader hose bins not located. of specified size and with smooth interior.	3.11.6.1.3
113a.	Through cabinet not of specified size, without shelf, or shelf not of specified size.	3.11.6.1.4
114a.	Any cabinet doors not flush, box-type, fully gasketed with full–length corrosion-resistant hinges, recessed two–point latching handles and flush, bolt-type locks.	3.11.6.1.4.1, 3.11.6.2.1, 3.11.1.6.2.2
115a.	Any cabinet illumination omitted or illumination not controlled through door switches.	3.11.6.1.4.2, 3.11.6.2.1, 3 .11.6.2.2
116a.	Any cabinet ventilation omitted.	3.11.6.1.4.3, 3.11.6.2.1, 3 .11.6.2.2
117a.	Attached equipment cabinet arrangement not as specified.	3.11.6.2
118a.	Equipment cabinet height, depth or recess less than specified; filler panels or skirts omitted; tops and fronts net flush horizontally and vertically. without gaps.	3.11.6.2.2
119a.	Pump operator’s panel not located, sized, labelled or arranged as specified.	3.11.6.3
120a.	Pump operator’s control installed without collars or plates.	3.11.6.3.1
121a.	Engine throttle control not screw-locking and center-pushbutton release type.	3.11.6.3.1.1
122a.	Heat exchanger control omitted from pump operator’s panel when heat exchanger is provided.	3.11.6.3.1.2
123a.	Foam liquid control not of type specified. without dial and pointer indicator, or without instruction plate.	3.11.6.3.1.3
1243.	Water tank shut-off control not linear or does not leave adjacent drain control.	3.11.6.3.1.4
125a.	Small suction inlet controls not located as specified or not linear.	3.11.6.3.1.5
126a.	Discharge valve controls not push-pull type, not symetrically grouped or not numbered as specified.	3.11.6.3.1.6
127a.	Fire pump drive control not linear and locking type.	3.11.6.3.1.7
128a.	Discharge relief valve control not rotary type; without dial and pointer indicator; or, without indicating light.	3.11.6.3.1.8
129a.	Discharge outlet drain controls not mounted beneath associated discharge valve control.	3.11.6.3.1.9

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Defect No.	Defect	Requirement Paragraph
130a.	Pump operator's gages installed without plates or collars.	3. 11.6.3.2
131a.	Pressure gages for each discharge outlet not of type specified; not properly sized; or, not grouped symmetrically with associated controls.	3. 11.6.3.2.3
132a.	Engine and transmission gages not provided, not sized and located or without indicator light as specified.	3. 11.6.3.2.4
133a.	Water tank gage not installed or not at eye-level.	3. 11.6.3.2.5
134a.	Pump operator 's instruments installed without plates or collars.	3. 11.6.3.3
135a.	Engine tachometer on pump operator's panel not interchangeable with tachometer on dashboard; not in proximity to throttle; or without means of checking,	3. 11.6.3.3.1
136a.	Ammeter missing, not properly sized or not located as specified on pump operator's panel.	3.11.6.3.3.2
137a.	Test connections missing, not properly sized or threaded, not located and labelled, or not plugged as specified.	3.11.6.3.3.3
138a.	Duplicate switch, and voltmeter for inverter missing or not installed as specified.	3.11.6.3.4
139a.	Electrical wiring for 110 Vac circuits not separate from 12 Vdc circuits.	3.11.7
140a.	Electrical equipment not in conformance with Requirement 1 of MIL-STD-454; not protected as specified: or, 6 V equipment.	3.11.7.1
141a.	Batteries not as specified.	3.11.7.2
142a.	Battery tray(s) not ventilated or treated; battery terminal guards omitted.	3.11.7.2.1
143a.	Battery circuits do not conform to the appendix to NFPA Code No. 1901.	3.11.7.3
144a.	Station battery charging provisions omitted or polarized receptacle not located as specified	3.11.7.3.1
145a.	Battery charging circuits on truck not protected with reverse polarity device or not switched as specified.	3.11.7.3.2
146a.	Truck operating lights not as specified.	3.11.7.4.1
147a.	Equipment operating lights not as specified.	3.11.7.4.2
148a.	Warning lights. do not conform to NFPA Code No. 1901 or are not as specified.	3.11.7.4.3

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Defect No.	Defect	Requirement Paragraph
149a.	Flood lights not as specified.	3.11.7.4.4
150a.	Floodlight receptacles not of type, location or wired as specified.	3.11.7.4.5
151a.	Lighting system controls not labelled in accordance with MIL-STD-1472.	3.11.7.5
152a.	Headlight control not of type specified.	3.11.7.5.1
153a.	Warning light controls not of type, location or not shielded to prevent glare.	3.11.7.5.2
154a.	Floodlight receptacle controls omitted,	3.11.7.5.3
155a.	Siren not of ttpe specified; not wired to connect to radio; not located and controlled; or without brake as specified.	3.11.8.1
156a.	Horns not of type, location, or controlled as specified.	3.11.8.2
157d.	Buzzer omitted or not switched as specified.	3.11.8.3
158a.	Handrails and grabrails not of size, mounting or locations as specified.	3.11.9
159a.	Stowed component attaching devices not positive locking, quick-release type.	3.11.10
160a.	Soft suction hose not of length, diameter or quality specified.	3.11.10.1
161a.	Reducer not of size, type, or not stowed as specified.	3.11.10.2
162a.	Ladders do not conform to NFPA Code No. 1931 or not of types specified.	3.11.10.3
163a.	Ladders not stowed where specified; not padded; or, not mounted in specified devices.	3.11.l0.3.l
164a.	Axes not of types specified, or not mounted where specified.	3.11.10.4
165a.	pike poles not of types or lengths specified or not stowed where specified.	3.11.10.5
166a.	Wheel chocks not of quantity specified or not stowed where specified.	3.11.10.6
167a.	Truck not serviced as specified or lubrication data missing.	3.12.1
168a.	Lubrication fittings not of type, location or accessible as specified.	3.12.1.1
169a.	Enclosures of integral lubricant reservoirs not as specified.	3.12.1.2
170a.	Fungus and moisture treatment not as specified.	3.12.2, 3.12.2.1
171a.	Truck not rustproofed in accordance with MIL-STD-1223, when specified.	3.12.3

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Defect No.	Defect	Requirement Paragraph
172a.	Overall dimensions of truck greater than specified.	3.14
173a.	Finish cleaning, treatment and painting not as specified.	3.17
174a.	Bolted connections not as specified.	3.19.3
175a.	Riveted connections not as specified.	3.19.4
176a.	Steel welding not as specified.	3.19.5.2
177a.	Aluminum welding not as specified.	3.19..3
	Minor	
201.	Color not as specified	3.16
202.	Identification marking omitted or not as specified.	3.18
203.	Castings and forgings not free of extraneous materials.	3.19.1
204.	Metal fabrication not as specified.	3.19.2
	Major, type T truck	
301.	Booster reel compartment does not have roll-up door or drain holes in bottom.	3.11.11.1.1
302.	Booster hose reel not of capacity or type specified.	3.11.11.1.2
303.	Booster hose not of type and couplings specified; or nozzle not of type or not stowed as specified.	3.11.11.1.5
304.	Booster system piping, valving, controls or outlet not as specified.	3 11.11.1.4
305.	Booster hose rollers not of material or not arranged as specified.	3.11.11.1.5
306.	Deck pipe not of type or in location specified.	3.1,11.2.1
307.	Deck pipe system piping, valving, controls or gage not as specified.	3.11.11.2.2
308.	Suction hose trays or tubes not of sufficient size, not properly located, or not of material specified.	3. 11.11.3.1
309.	Hard suction hose of wrong length, diameter, or couplings not of type specified.	3.11.11.3.2
310.	Strainer not of size, type or not stowed as specified.	3.11.11.4
311.	Warning light does not conform to NFPA Code No. 1901; or not of type, location or controls as specified.	3.11.11.5

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Defect No.	Defect	Requirement Paragraph
	Major type II truck	
401.	Hydraulic system not of materials compatible with nydraulic nils specified.	3 .11.12.1
402.	Hydraulic fittings not of type specified for port size applicable.	3 .11.12.1,1
403.	Hydraulic lines not installed as specified.	3.11 .12.1.2
404.	Pressure hoses not of type or sizes specified; not pressure rated to or above relief setting; or bent beyond minimum radius.	3.11.12.1,2.1
405.	Pressure tubing not of type or sizes specified; burst pressure less than specified; or improperly installed.	3.11.12.1.2.2
406.	Pressure tube fittings have a burst pressure less than specified,	3.11.12.1.2.3
407.	Suction lines without flexible characteristics or hose and clamps not as specified.	3.11.12.1.2.4
408.	Return lines not as specified,	3.11.12.1.1.5
409.	Hydraulic system filters omitted.	3.1l.12.l.4
410.	Return line and pressure line filters not remove-able without spilling or draining oil, or not equipped with provisions for remote contamination indication,	3.11.12.1.4.1, 3.11.12.1.4.2
411.	Hydraulic fluid reservoir not designed, constructed or installed as specified.	3.11.12.1.5
412.	Pressure relief valve not set to maximum system pressure.	3.11.12.1.6
413.	Hydraulic cylinders not manufactured as specified.	3.11,12.1.7
414.	Turntable not of dimension specified.	3.11.12.2
415.	Boom and ladder does not have nozzle recess or is not of dimensions specified.	3.11.12.3
416.	Water tower water system valves not of type or material as specified.	3.11.12.4
417.	Master control valve not of type or material specified.	3.11.12.4.1
418.	Water tower inlets not plugged as specified.	3.11.12.4.2
419.	Inlet drains not as specified.	3.11.12.4.3
420.	Remote control nozzle not of size specified; does not have vertical and horizontal sweep functions separate; or; does not have duplicate controls as specified.	3.11.12.5

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Defect No.	Defect	Requirement Paragraph
421.	Outriggers not equipped with automatically activiated flashing lights.	3.11.12.6
422.	Controls, gages and switches on control panel not arranged and labelled as specified.	3.11.12.7
423.	Tower hydraulic control not one-handed type.	3.11.12.7.1
424.	Control handle of master control valve not located on panel as specified,	3.11.12.7.2
425.	Water pressure gage not of range and size specified.	3.11.12.7.3
426.	Remote nozzle and stream shaper controls not mounted together on panel.	3.11.12.7.4
427.	Outrigger control not equipped with lockout or unlocking switch not protected as specified.	3.11.12.7.5
428.	Master electric switch not equipped with indicator light.	3.11.12.7.6
429.	Communication system between control station and top of boom omitted.	3.11.12.8
430.	Warning lights do not conform to NFPA Code No. 1901; not dual: not of type, location or controlled as specified.	3.11.12.10

4.5.2 Tests.

4.5.2.1 Maintainability test. After all tests involving the use of the truck engine have been completed, perform the maintenance operations specified in 3.6a through 3.6i. The engine-driven belts, alternator, coolant hoses, battery cables, batteries, and starter shall be inspected upon removal and shall be reinstalled instead of being replaced if inspection indicates a satisfactory operating condition of the components involved. Screens and strainers shall be cleaned and inspected, and may be reinstalled, if inspection indicates a satisfactory operating condition of the screen or strainer. Fluids and filters shall be replaced. Any maintenance operation performed that exceeds the time allocated in 3.6 for that operation, including inspection and cleaning of the component involved , shall constitute failure of this test.

4.5.2.2 Mobility test, Prepare the truck for testing as follows:

a. Fill the water tank with not less than 500 gal of water.

b. Fill the foam liquid tank with not less than 60 gal of foam liquid conforming to MIL-F-24385.

c. Fill the fuel tank to capacity with diesel fuel.

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d. Fill all lubricant and hydraulic reservoirs. Lubricants shall be those indicated on a lubrication data plate or lubrication decals.

e. Load the truck with an evenly distributed simulated load of 1500 lbs.

f. Load the truck with a driver, a passenger and simulated dead-weight of not less than 175 lbs each in each jump seat and on the rear platform. If the driver's and passenger's collective weights do not equal or exceed 350 lbs, a deadweight load equal to the difference between their collective weights and 350 lbs shall be placed on the cab floor on the passenger's side. Secure the simulated loads by a positive means that will prevent load shifting or loss of any part of the load during running, turning or braking during the test. When the truck is fully prepared for testing, start the engine and allow it to operate for a sufficient time-span to bring the engine coolant temperature up to the engine manufacturer's recommended operating temperature. When engine coolant temperature rise is sufficient, operate the truck over a test course of 100 miles of paved high-way and 50 miles of graded dirt road. The test course shall be negotiated at maximum posted speeds up to 55 mph on paved high-way and up to 30 mph on graded dirt road. Nonconformance of the truck to any performance or mobility requirement of NFPA Code No. 1901 shall constitute failure of this test.

4.5.2.3 Pumping capacity test. The pumping capacity test conditions arc an ambient air temperature of +125° F and altitude of 2000 ft above sea level. These test conditions may be actual or simulated. Simulated test conditions shall be approved before testing by the contracting officer or the designated repre-sensitive. The test shall consist of testing the priming device operation, the suction of the pump, and 3 hours of actual pumping time. The truck shall be prepared by attaching 20 ft of 6 inch suction hose to the 6 inch suction inlet on the front of the truck. The suction hose shall consist of two lengths of the hose specified in 3.11.11.3.2 1.3.2 and shall be terminated with a strainer as specified in 3,11.11,4, Place the strainer end of the hose in a reservoir or similar body of water that will permit the withdrawal of not less than 5,000 gal. without drafting mud or silt. Attach a calibrated suction gage and a calibrated pressure gage equivalent to the gages specified in 3.11.6.3.2.2 and 3.11.6.3.2.1 respectively, to the test connections on the pump operator's panel. Attach hoses to outlets on the truck and direct their flows back to the water source, Fill the fuel tank to its maximum capacity. Start the truck engine, engage the pump drive and bring the engine up to the recommended rpm speed for priming, that is recommended by the priming device manufacturer, Operate the priming device and observe the test suction gage to determine that suction has been established.

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Increase engine speed and open the outlet control valves until the truck is discharging 1000 gpm at 150 psi as observed on the calibrated discharge test gage, Continue pumping at this rate for not less than two hours. After two hours, adjust outlet control valves and engine speed to pump 700 gpm at 200 psi as observed on the calibrated discharge test gage. Continue pumping at this rate for not less than 30 minutes. After 30 minutes, adjust outlet control valves and engine speed to pump 500 gpm at 200 psi as observed on the calibrated discharge test gage. Continue pumping at this rate for another 30 minutes. Failure of the priming device to operate or establish suction within 10 seconds, or failure of the truck to establish and maintain the discharge rates at the pressures specified shall constitute failure of this test.

4.3.2. 3.1 Pump drive temperature test. During the pumping capacity test specified in 4 .5.2.3, a thermocouple and calibrated temperature readout device shall be connected into the pump drive gear oil reservoir. The gear oil temperature shall be monitored on the readout device throughout the entire 3 hours of the pumping capacity test. A gear oil temperature that exceeds the pump drive manufacturer's recommended maximum gear oil temperature shall constitute failure of this test.

4.5.2 .3.2 Discharge relief valve test. At the end of the pumping capacity test, establish a discharge rate of 1000 gpm at 75 psi pressure. Raise the discharge relief valve setting to not more than 85 psi. Slowly close the control valves on the discharging lines while observing the test pressure gage. Open the control valves on-the discharging lines and reset the pressure relief valve to 160 psi. Increase the discharge-pressure to 150 psi while maintaining a discharge rate of 1000 gpm. Slowly close the control valves on the discharging lines while observing the test pressure gage. A pressure reading increase of more than 30 psi above the discharge pressure established before the control valves on the discharging lines were started to close, throught full closing of the valves, shall constitute failure of this test.

4.5.2.4 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 7. As a minimum, noise levels shall be measured when equipment is operating under fulll load. MIL-STD-1474, paragraph 5.1.2.1.4 contours shall be taken at not fewer than 12 equal (horizontal) arc increments, one increment shall include data from the noisiest position. Additionally, the noise level at the typical operating position shall be provided as dB(A) level. Failure to comply with MIL-STD-1474 provisions shall constitute failure of this test.

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4.5.2. 5 Environmental tests.

4.5. 2. 5.1 Altitude pumping test. Any truck ordered for a service altitude over 2000 ft above sea level, shall be tested as specified in 4.5.2.3, except that the test conditions shall be an ambient air temperature of 75°F. + 10°F and an altitude equal to the service altitude ordered. Failure of the test shall be as specified in 4.5.2.3.

4.5.2.5.2 Cold starting and operating test. Cold soak the truck for a period of not less than 24 hours in an ambient air emperature of-25°F. Start the truck and begin operating (driving) it within 30 seconds of start. Any failure of the engine to to start within 30 seconds, engine stalling after commencement of operation, or failure of the transmission to shift into appropriate gear ranges shall constitute failure of this test.

4.5.2.5.3 Hot starting test. Conduct the test in an ambient air temperature of -125°F Start and operate the truck engine until the engine coolant temperature is raised to the engine manufacturer's recommended operating temperature. Strip the engine and wait for not less than 2 minutes. After the 2 minute wait, restart the engine. Failure of the engine to restart shall constitute failure of this test.

4.5.2.6 Fuel system capacity test. Measure the fuel tank capacity. Before starting the pumping test specified 4.5.2.3, fill the fuel tank to capacity and record the elapsed time indicated by the engine hour meter to the tenth of an hour. Upon completion of the pumping test, refill the fuel tank to capacity. Record the gal required to refill the tank to the nearest tenth of a gal and record the time indicated on the engine hour meter to the tenth of an hour. Subtract the pre- test record engine hour time from the post–test recorded engine hour time. Divide the recorded gal required to refill the tank by the remainder, to obtain the average gal per hour used. Acid the average gal per hour used to the number of gal required to refill the tank to obtain a sum which represents the fuel tank capacity. Nonconformance to 3.11.1.2.3 shall constitute failure of this test.

4.5.2.7 Brake tests.

4.5.2.7.1 Service brake test. During or immediately after the mobility test specified in 4.5.2.2. and while the truck is still loaded, drive the truck up onto a grade that is not less than 30 percent. When the truck is completely on the grade, bring it to a full stop on the grade and hold it there with only the service brake for one minute. Release the service brake and drive the truck onto a paved, dry, level surface that is free of loose material. Bring the truck speed to 20 mph and apply the service brakes in a manner that simulates an emergency stop. Measure the distance between the point where the service

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brakes were applied and the point where the truck stopped. then resume driving. Drive the truck around to where it is running down the same grade it had been driven up earlier in the test, and bring it to a full stop on the grade. Hold the truck on the grade with only the service brake for one minute. Any drifting of the truck from the full stop position, either up or down while stopped on the grade, or failure to come to a full stop within 30 ft on the level surface while applying no more than 75 lbs of force on the brake pedal, shall constitute failure of this test.

4.5.2.7.2 Parking brake tesst. Connect a calibrated air pressure gage to the truck's air tank. Start the truck engine and bring the tank's air pressure up to not less than 100 psi. Pull the parking brake completely "on" and leave the control in its full extended position. Observe and record the air pressure reading on the calibrated pressure gage. Leave the parking brake "on" for not less than 24 hours. After 24 hours, observe and record the air pressure reading on calibrated air pressure gage. A drop in air pressure in the air tank of 25 or more lbs after 24 hours or a force of greater than 50 lbs required to fully apply the parking brake, shall constitute failure of this test.

4.5 .2.7.3 Engine brake test. Operate the truck over a 5–mile-long segment of paved highway at posted speeds up to 55 mph. At two different intervals, turn the engine brake switch to the "on" position. During the intervals when the switch is "on", remove pressure from the accelerator and observe if the truck speed is immediately retarded. Between the intervals when the switch is "off", remove pressure from the accelerator and observe if the truck speed is immediately retarded. Failure to immediately retard truck speed when pressure is removed from the accelerator while the engine brake switch is in the "on" position, or any immediate retarding of the truck speed that is not attributable to normal engine compression when the engine brake switch is in the "off" position, shall constitute failure of this test.

4.5.2.8 Generating system tests.

4.5.2.8.1 Alternator output test. Start the trurk engine and increase the engine speed to 1000 rpm. Use an ammeter to measure the alternator/ voltage regulator output in amps at this engine speed. Then slow the engine to the engine manufacturer 's recommended idle speed and use the ammeter to measure the inverter output in amps at the idle speed. An alternator/voltage regulator output of less than 70 amps at an engine speed of 1000 rpm or an inverter output of less than 17.40 amps at engine idle speed shall constitute failure of this test.

4.5.2.8.2 Inverter voltage control test, Measure the inverter voltage and cyclic output under inductive loads of 2000, 4000 and 6000 W. Any voltage reading that is less than 105 Vac or more than 115 Vac, or any cyclic output that varies by more than 1 percent from 60 Hz, shall constitute failure of this test.

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4.5.2.9 Water tank pumping test . Fill the water tank with not less than 500 gal of water. Connect hoses to a sufficient number of discharge outlets to accommodate a flow of not less than 500 gpm. Open the appropriate discharge valves. Start the engine. engage the pump drive. prime and increase engine speed to at least 2200 rpm. Open the water tank shut-off valve in one stroke. Observe the water tank water level gage and note the time required to empty the water tank. When the water tank water level gage indicates the tank is empty, close the water tank shut-off valve in one stroke. Remove the cover of the water tank and determine if all the water has been pumped from the tank. Failure of the truck to empty the water tank in 1 minute, or less, shall constitute failure of this test.

4.5.2.10 Foam system test. Fill the foam liquid tank to capacity, using either foam liquid or a dyed-water substitute. Connect a nose to a discharge outlet that will accommodate a flow of not less than 200 gpm. Open the appropriate discharge valve. Connect a suction those to an apropriate suction inlet. Start the engine, engage the fire pump drive, prime, and establish suction and a discharge flow of not less than 200 gpm. Set the foam liquid metering valve or proportioned to deliver a 3 percent solution. Open the foam liquid shut-off valve for exactly 1 minute and then close the valve. Refill the foam liquid tank to capacity. Measure and record the amount of liquid required. Reset the foam liquid metering valve or proportioner to deliver a 6 percent solution. Open the foam liquid shut-off valve for exactly 1 minute and then close the valve. Refill the foam liquid tank to capacity. Measure and record the amount of liquid required Drain the foam liquid tank and flush and clean the entire foam system. Failure of the foam system to have discharged not less than 6 gal of liquid in solution in 1 minute at the 3 percent setting or not less than 12 gal of liquid in solution in 1 minute at the 6 percent setting shall constitute: failure of this test.

4.5.2.11 Hydrostatic test. Prepare the truck for esting by emptying the water and foam liquid tanks and cleaning, if necessary, the foam systen piping and controls. Cap or plug all suction inlets and all discharge outlets on the entire truck with valves that will facilitate bleeding of air from the fire pump. Discharge lines to either a deck pipe or a remote control nozzle may be plugged by closing the deck pipe or nozzle control. A booster hose shall be plugged by closing the ball-shutoff at the nozzle. Close all drain valves and fully ope all control valving between the fire pump and the suction inlets and discharge outlets. Connect calibrated pressure gages to both female test connections on the pump operator's panel. Connect an external water pumping system to one of the small suction inlets. Start the external pumping system and charge the fire pump and piping system while bleeding all air out of the system. Bring the water pressure in the fire pump up to not less than 350 psi on both gages and maintain that pressure for not less than 10 minutes. Inspect the fire pump case, manifolds, control and drain valves, and all piping for evidence

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of leaks. Inspect the interior of the water and foam liquid tanks for evidence of water indicating leakage past the check valves. Any evidence of leakage in or around the pump, manifolds, control Or drain valves, piping, booster hose and fittings or check valves shall constitute failure of this test.

4.5.2.12 Hose bin and cabinet bottom test. Remove the floors from eachn hose bin and cabinet before testing. With the floor removed , chalk-mark a square ft area on the bottom that does not incorporate a crossmember. Place a 1 inch thick, one square foot piece of wood on the chalk-marked area. Place a static load of not less than 200 lb on the piece of wood. After 10 minutes, remove the test load and wood piece. Examine the bottom for cracks or deformation that does not return to original flatness when the load is removed. Evidence of fracture cracks or any permanent deformation shall constitute failure of this test.

6.5.3 EMI test. When conformance to EMI emission and susceptibility has been specified (see 6.2), the truck shall be tested in accordance with the test requirements of SAE J551. Nonconformance to SAE J551 shall constitute failure of this test.

4.5.4 Type I truck test.

4.5.4.1 Deck pipe line flow test. Prepare the truck for testing as specified in 4.5. 2. 3, except that the ambient air temperature may be any temperature above +32°F: the altitude may be any altitude above sea level; and, the discharge shall be through the deck pipe line only. Open the deck pipe nozzle and the deck pipe control valve. Start the truck engine. engage the pump drive, prime the pump and establish suction. Adjust engine speed and control valve opening to establish a flow of 750 gpm at 150 psi through the deck pipe. Failure to establish a flow of 750 gpm at 150 psi with an engine speed that is not more than the engine manufacture's recommended maximum rpm shall constitute failure of this test.

4.5.5 Type II truck tests.

4.5.5.1 Pumping stability test. Prepare the truck for testing as specified in 4.5.2.3, except that the ambient air temperature may be any temperature above +32°F; the altitude may be any altitude above sea level; and the discharge of water shall be through the remote control nozzle on the end of the boom only. Use the truck engine to power the hydraulic system and extend the outriggers until they are locked into position to stabilized the truck. Elevate the boom from its riding position to an angle of 10° above the horizontal and extend it to its maximum extended condition. Rotate the boom to a heading, of 35° off the heading of the truck and depress it to 10° below the horizontal. Engage the fire pump drive, prime and establish suction. Open the master control valve and adjust engine speed and the remote control nozzle until a flow of 1000 gpm.

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at 150 psi is discharging thru the remote control nozzle in ahosizontal direction away from the truck. Release the hydraulic controls for not less than 2 minutes. After 2 minutes, elevate the boom to as nearly vertical as possible and simultaneously adjust the elevation of the remote control nozzle to continue the discharge in a horizontal direction away from the truck. Release the hydraulic controls while the boom is in this elevation for not less than 2 minutes. Repeat the cycles of depressing and elevating the boom. with release of the hydraulic controls at the depressed and elevated angles specified, on headings of 90° and and 140° from the heading of the truck. The discharge rate shall be maintained continuously throughout all movements of the boom. After the last cycle, discontinue discharging by closure of the master valve and lowering of engine speed. Depress the fully extended boom to horizontal while still on the 1400 heading away from the truck heading. Secure a 300 lb load on the ladder of the boom in a position most likely to be occupied by personnel operating the remote control nozzle with the controls near the nozzle. Elevate the boom and load to an angle of 45° above the horizontal. Open the master control valve, adjust engine speed and the remote control nozzle to reinstate a discharge of 1000 gpm at 150 psi through the nozzle in a horizontal direction away from the truck. Release the hydraulic controls for not less than 2 minutes and then elevate the boom and adjust the remote control nozzle to attain an elvation of 75° above the horizontal with the discharge in a horizontal direction from the truck. Release the hydraulic controls for not less than 2 minutes. Repeat these cycles of depressing the boom to 45° and elevating to 75° on headings of 90° and 35° off the heading of the truck, with the discharge rate maintained continuously throughout the movements. of the boom Failure of the outriggers or truck tires to remain in contact with the ground. or failure of the boom to maintain elevation and heading within +5° of either, when the hydraulic controls are released, shall constitute failure of this test.

4.5.5.2 Ladder stability test. Position the truck 18 ft from the side of a solid structure of sufficient height and strength to support a loaded. leaning boom. Use the truck engine to power the hydraulic system and extend the outriggers until they are locked into a position to stabilize the truck. Elevate the boom from its riding position to an angle of 10° above the horizontal and extend it to its rated extended length. Rotate the boom horizontally away from the structure until the boom heading is 90° off the truck heading. Secure a 400 lb load to the end of the boom and elevate the boom to an angle of 45° above the horizontal . Rotate the boom to a heading of 35° off the truck heading; then stop. and rotate back to a heading of 140° off the truck heading. Return the boom to the 90° heading off the truck heading and release control of the hydraulics for not less than 1 minute. Depress the boom to a horizontal elevation and retract the extended boom. Secure an additional 400 lb load to the end of the boom and then elevate the boom to 45° above the horizontal. When the boom reaches the elevation of 45° above the horizontal, extend the boom to its maximum extended length. Then elevate the boom to an angle of 75° above the horizontal and rotate it to headings of 35° and back to 140° off the truck heading. Return to the 90° heading off the truck heading and release control of

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the hydraulics for not less than 1 minute. Depress the boom to an angle of 45° above horizontal and repeat the rotations to headings of 35° and 140° off the truck headings. Then return to the 90° heading and release control of the hydraulics for not less than 1 minute. Retract the boom and depress it to the horizontal. Remove the secured loads. Extend the boom to its maximum extended length and elevate it to its maximum height. Rotate it 180° toward the structure and depress it until the boom rests squarely against the structure. Secure 1600 lb of load evenly over the entire length of the ladder portion of the extended boom. Leave the loads on the ladder for 2 hours. Then remove the loads and inspect the ladder and boom for evidence of fracture or permanent deformation. Failure of the outriggers or truck tires to remain in contact with the ground, failure of the boom to maintain elevation and heading with ± 5° of either when the hydraulic controls are untouched, or evidence of either cracking or permanent deformation shall constitute failure of this test.

4.5.5.3 Hydraulic system tests.

4.5.5.3.1 Hydraulic back-up test. With the truck engine off, turn the master electric switch at the control station panel to the "on" position to stabilize the truck. Use the one-handed tower control to elevate, rotate, extend. depress and retract the boom with hydraulic power supplied only by the electric back-up motor. The operation of the boom shall contiune for a period of not less than 5 minutes. Within the 5 minute test period. the boom shall be fully elevated and fully depressed 2 time, rotated ompletely through 360° rotations in both drections 2 times, and shall be fully extended and fully retracted 2 times. Failure of the back-up motor to supply sufficient hydraulic power to operate the boom through all of the elevations, depressions, rotations, extensions and retractions within the test period shall constitute failure of this test.

4.5.5.3.2 Pump inlet pressure test. Install a pressure (vacuum) gage in the pump suction line immediately adjacent to the pump and installed a thermocouple in the reservoir. Heat the hydraulic fluid to 140°F. + 10° F. Start the engine and bring it up to maximum governed speed. Operate the water tower while observing the gage and oil temperature. A pressure drop below the pump manufacturer's rating while the hydraulic fluid is at the test temperature and the engine is operating at maximum governed speed shall constitute failure of this test.

4.5.5.3.3 Hydraulic pump contaminent tolerance. Compare the pump manufacturer's B_{10} contaminent tolerance to the filter B_{10} curve. A contaminent tolerance of the pump that is less than the level of the filter at any point on the B_{10} curve, shall constitute failure.

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4.5.5.3.4 Hydraulic filter test. The test may be performed on a hydraulic test stand in place of an on-vehicle test. Connect a pressure gage, temperature gage and a flowmeter into the filtered (return or pressure) line. Pressurize the line to 100 psi. for return line filters or 1.5 times the maximum system pressure for pressure line filters. Hold the pressure for not less than 1 minute and then examine for evidence of leaks. Remove the pressure from the line and install another pressure gage on the other side of the filter and a new filter element in the filter housing. Establish a maximum flow through the filter at an oil temperature of 150°F, + 5° F. Maximum flow through the filter shall be the predetermined flow occuring when the boom is depressed while the engine is operating at maximum governed engine speed. Measure and record the pressure drop across the filter assembly. Remove the pressure from the line, and plug the filter element chamber. Re-establish and maintain an oil temperature of 150° F, + 5° F. Increase upstream pressure to 10 psi. above the recorded pressure drop across the filter assembly. Observe the contamination indicator and increase. upstream pressure until the indicator reflects that the filter is bypassing, Record the pressure req~li.reel for bypass opening. Then continue to increase upstream pressure until maximum flow is attained through the bypass valve. Observe and record this pressure. Nonconformance of return line filters to 3. 11.12.1.4.1 or pressure line filters to 3. 11.12.1.4.2 shall constitute failure of this test.

4.5.5.3.5 Hydraulic cylinder proof test. The test shall be performed on a hydraulic test stand. Secure the cylinder and extend the piston rod to its maximum extended position. Fill the cylinder through the head end port with oil conforming to MIL-L-2104, grade 10 or MIL-L-46167 and open the rod end port . Apply pressure through the head end port with the same oil until the pressure is 1.5 times the pressure relief valve setting. Hold this pressure for not less than 1 minute. Release the pressure and drain the oil from the cylinder while leaving the head end port open. Retract the piston by filling the cylinder with the same oil through the rod end port. Apply pressure through the rod end port with the same oil until the pressure is 1.5 times the pressure relief valve setting. Hold this pressure for not less than 1 minute. Any evidence of rupture, permanent deformation, internal or external damage shall constitute failure of this test.

4 .5.5.3.6 Hydraulic system cleanliness test. Use a liquid automatic particle counter to record particle counts, The particle counter shall be calibrated in accordance with ANSI B93.28, using “AC” Fine Test Dust. Start the truck engine and bring it up to no-load governed speed. Activate the hydraulic pump drive and manipulate all hydraulic controls through full stroke so that each hydraulic circuit has had oil pumped through it for not less than 5 minutes. Connect the calibrated counter into the hydraulic system, upstream of the filter(s) used for cleanup or any flow restricting devices, in a return line having a flow rate which is not less than 5 times the flow rate of the calibrated counter. Connection of the counter shall be via a bypass line and a branch line. The bypass

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line may return oil directly to the reservoir. The branch line shall tee into the bypass line. shall be connected to the counter on the other end, and shall be not more than 12 inches long. Restart the engine, engage the hydraulic pump drive and establish a truck engine speed of not less than 1000 rpm. Manipulate the controls through full stroke so that each hydraulic circuit has had oil pumped through it for not less than 5 minutes. Record the maximum particle counts indicated by the counter as each circuit is opened. Any particle count that is greater than the particle counts specified in 3. 11.12.1.8 shall constitute failure of this test.

4.5.5.3.7 Hydraulic external leakage test. Observe and record accumulated leakage at practical intervals over a 12 hour period following all tests involving use of the hydraulic system. Leakage in excess of the limits specified in 3.11.12.1.9 shall constitute failure of this test.

4.5.5.3.8 Turntable rotation test. Using the truck engine to power the hydralic system, extend end the outriggers until they are locked in a position to stabilize the truck. Elevate the boom from its riding position to an angle of 10° above the horizontal. Rotate the turntable 10 times in a clockwise direction; stop, and then rotate it 10 times in a counter-clockwise direction. Failure of the turntable to make continuous 360° turns in either direction shall constitute failure of this test.

4.5.3.3.9 Throttle control switch test. During the test specified in 4.5.5.1, make minor adjustments to the headings and elevations of the boom just prior to relinquishing hydraulic control of the boom position. Activate the the throttle control switch while making the minor adjustments to headings and elevations. Any surging of the truck engine in response to hydraulic control movements while the throttle control switch is activated shall constitute failure of this test.

4.5.5.6 Tread plate load test. Unlatch the tread plate at the pump operator's panel and pull it all the way out to the operating position. Load a test dead - weight of not less than 300 lbs on the outermost edge of the tread plate. Remove the load and return the tread plate to its retracted position beneath the running board. Unlatch and pull the thread plate all the way out a second time. Failure of the threadplate locking mechanism to latch the treadplate in the retracted position, or binding of the sliding mechanism supporting the treadplate after being subject to the test load, shall constitute failure of this test.

4.6 Inspection of packaging. The preservation. packing and marking of the truck and equipment shall be examined to determine compliance with the quality assurance provisions of MIL-STD-281 and MIL-E-10062. as applicable.

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5. PACKAGING

5.1 Preservation. The degree of preservation shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A. Each complete truck, with equipment, shall be preserved in accordance with MIL–STD–281 , level A (mobile), with the following additions and exceptions.

5.1.1.1 Engine. Engine and engine accessories shall be preserved in accordance with MIL-E-1OO62, type I, method I, level A.

5.1.1.2 Cooling system. The cooling system shall be preserved by the water and antifreeze procedure as specified in MIL-STD-281, except the antifreeze shall conform to MIL–A-46153.

5.1.1.3 Equipment. Equipment shall be secured as provided for, except exposed equipment subject to pilferage, damage or loss shall be prepared as “on vechicle equipment” in accordance with MIL-STD-281.

5.1.2 Level E. Each complete truck shall be preserved in accordance with MIL-STD-281, level B (mobile), and with the additions and exceptions as specified for level A.

5.1.3 Level C. Each complete truck and all equipment issued with the truck shall be sutiably preserved to afford protection against deterioration and damage from the contractor to the initial destination. In addition, the following specific requirements shall be accomplished. Items with critical surfaces required for fit or function shall be provided protection by the use of preservative coatings. volatile corrosion inhibitors, or desiccated packs. Items requiring protection from physical damage to surfaces, or which are fragile by nature, shall be protected by wrappings, pack compartmentization, or carbonizing of the individual item. Exposed equipment subject to easy pilferage or loss shall be placed in secured compartments, truck cab or in wooden boxes. Wooden boxes, when used, shall be secured to the truck with metal strapping and in a manner to prevent damage to the painted surfaces of the truck. The cooling system shall be filled with a fresh clean solution of 50 percent water and 50 percent antifreeze conforming to MIL-A-46153 and a tag attached indicating the temperature to which the cooling system can be subjected before damage. The crankcase and all other lubrication reservoirs shall be filled to the operating level with the grade of military lubricant designated for use in the temperature range at the shipping destination. A tag shall be attached in a conspicuous location to indicate which military lubricant has been used and its grade and temperature range. A deprocessing guide shall be furnished indicating in detail the preservation material to be removed and other servicing required prior to placing the truck in operation.

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5.2 Packing (see 6.7). Each complete truck, preserved as specified in 5.1, shall not require packing.

5.3 Marking (see 6.7). Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The type I truck is intended for use in fighting fires in low structures in Military operational areas. The type II truck is intended for use in fighting fires in multistory structures in Military operational areas.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Type of truck required (see 1.2).

c. When a first article is required for inspection and approval (see 3.2).

d. When the truck shall be capable of pumping at altitude, and the service altitude required (see 3.10.1).

e. When a radio communication system shall be provided (see 3.11.2.1.8).

f. When the truck should be rustproofed (see 3.12.3).

g. When EMI emission and susceptability characteristics are not required (see 3.13).

h. When the color of the truck should be other than gloss lime-yellow (see 3.16).

i. When exrerior markings on the truck are required, and the marking data to be used (see 3.18.1).

j. When welding process to be used will be specified by the Government (see 3.19.5.2).

k. Level of preservation required (see 5.1).

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6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1644) and delivered in accordance with the approved CRDL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1413 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited by the following paragraphs:

Paragraph No.	Data requirements	Applicable DID No.	Options
3.11.4, 3.11.4.3.1, 3.11.5, 3.11.7	Commercial Manual	DI-M-4022	Separate lubri-eating instructions and charts are required
3.19.5.1	Certification, welders qualification	DI-H-24029	

6.3 First article. When a first article inspection is required, the item should be a preproduction model. The first article should consist of one truck. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the first article inspection results and disposition of the document's first article.

6.4 Definitions.

6.4.1 Ambient temperature. The temperature of the atmosphere measured at the intake side of the engine cooling radiator.

6.4.2 Automatic type transmission. A Transmission in which normal gear effected automatically wiwthout manual assist.

6.4.3 Excessive surging. Any movement of the tank contents that would cause the driver of the truck to lose control of the truck during acceleration, deceleration, or any turn.

6.4.4 Flattening. Throughout the bend of a tube, the maximum diameter of the tube shall be equal to or greater than the tube's nomimal diameter, and the tube's maximum diameter minus the tube's minimum diameter shall be not greater than 15 percent of the nominal tube diameter.

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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6.6 Lubricants. The contracting officer should furnish a list of Military lubricants applicable to the truck covered by this specification as contained in Federal Supply Catalog, Department of Defense Section, Identification List C9100-IL, for FSC Croup 91.

6.7 Packing and marking. Levels of packing have not been included as they are not appropriate for this item. For purposes of preservation/packing level marking, the marking shall be as required by the customer,

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-0010224M(ME)	Trucks, Fire Fighting: Structural, 1000 GPM

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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